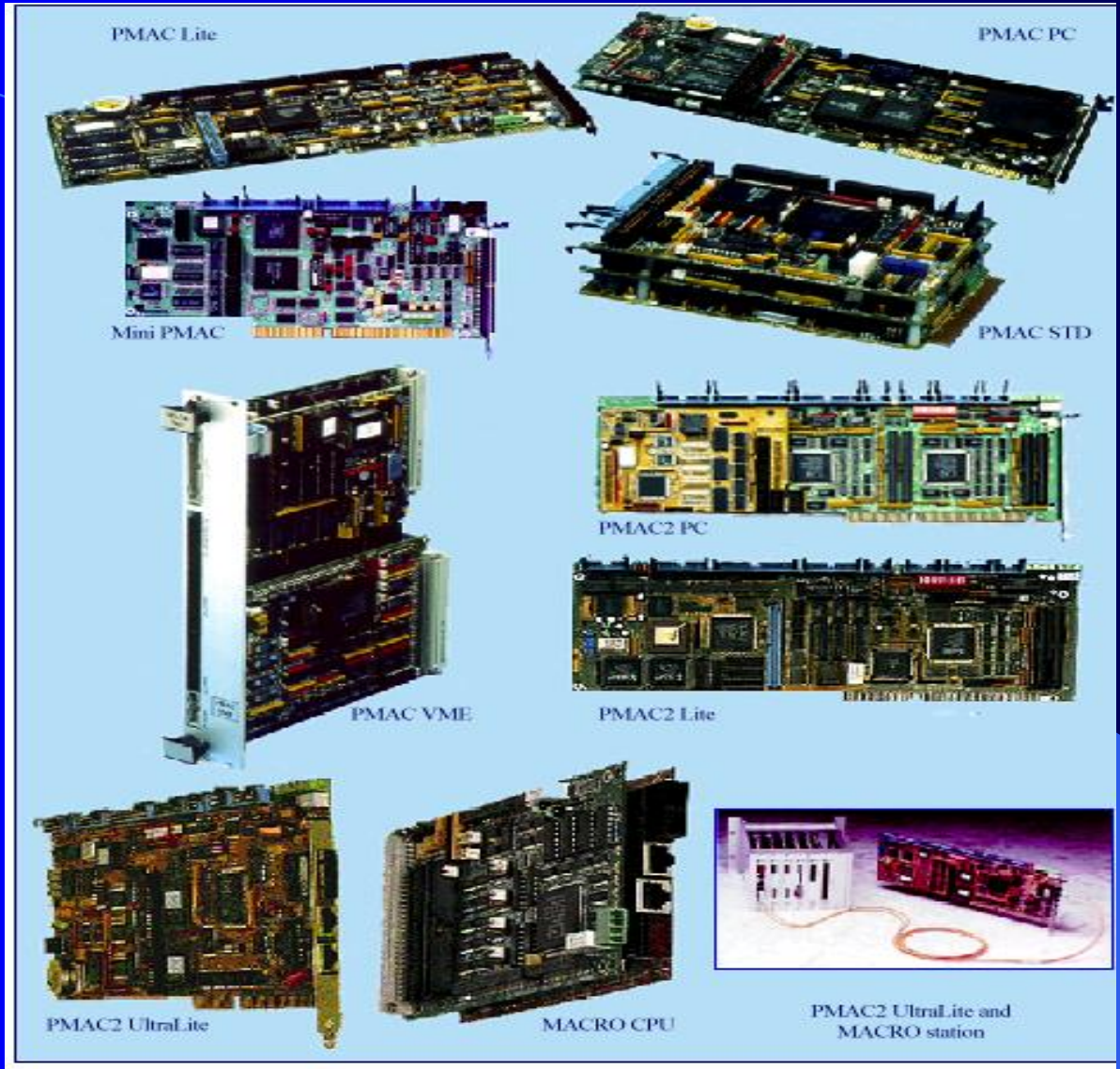
E-MOTION

PMAC多轴运动控制卡（中级培训）



LINEAR 插补方式

对PMAC 用户直线插补是最通用的方式. 直线插补轨迹包括加减速时间和运动时间.

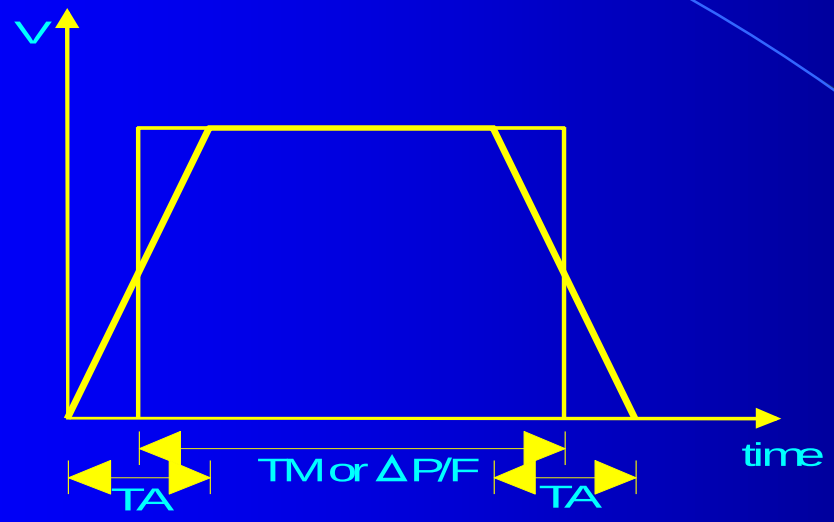
需要学习:

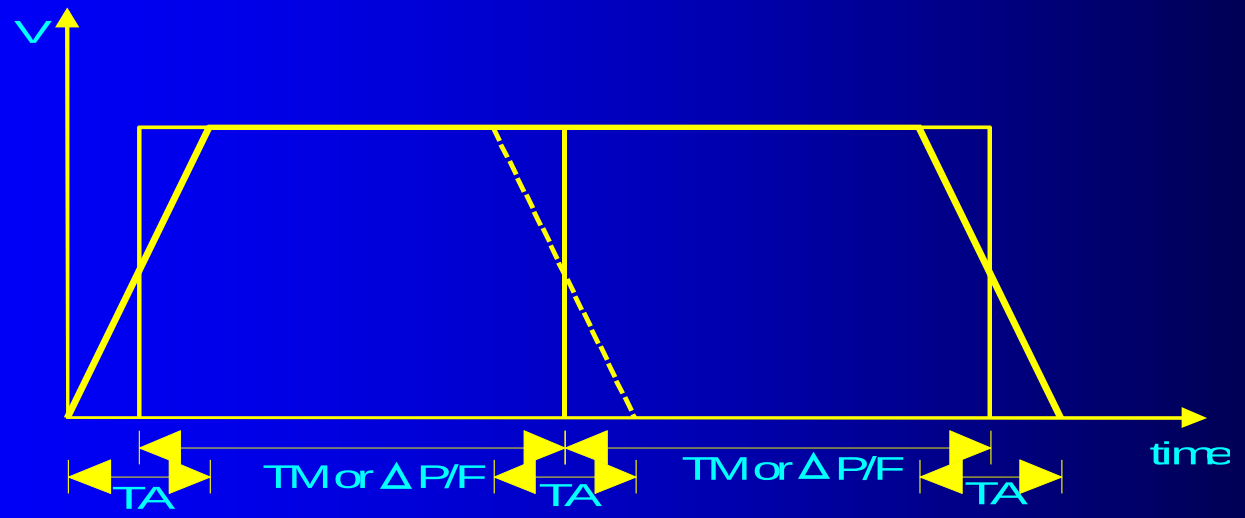
什么是直线插补模式的轨迹

如何改变轨迹的特征

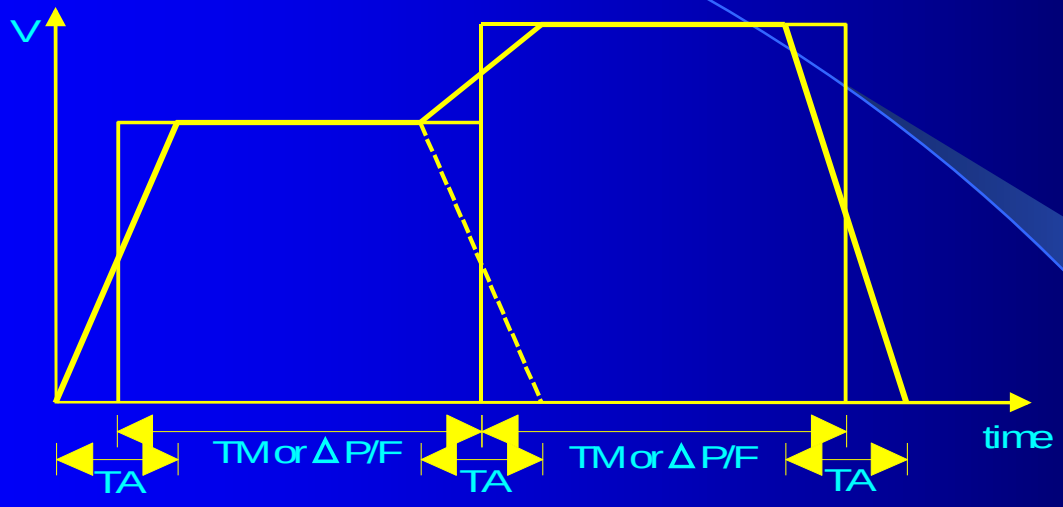
如何实现多个运动的速度混合

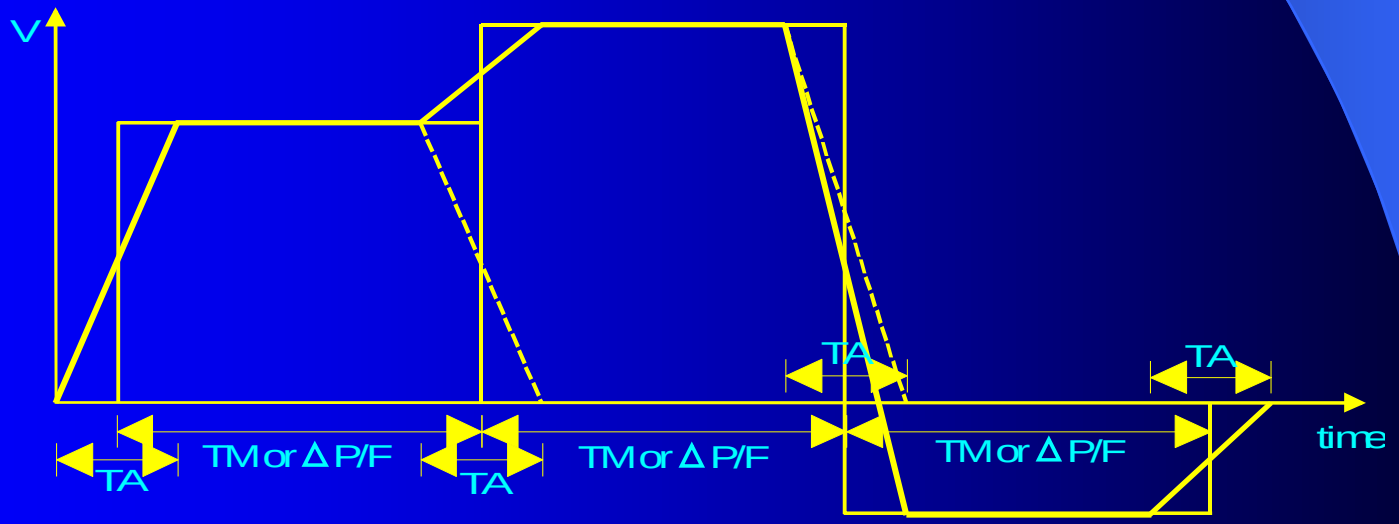
Linear 模式的轨迹
小Acceleration 时间





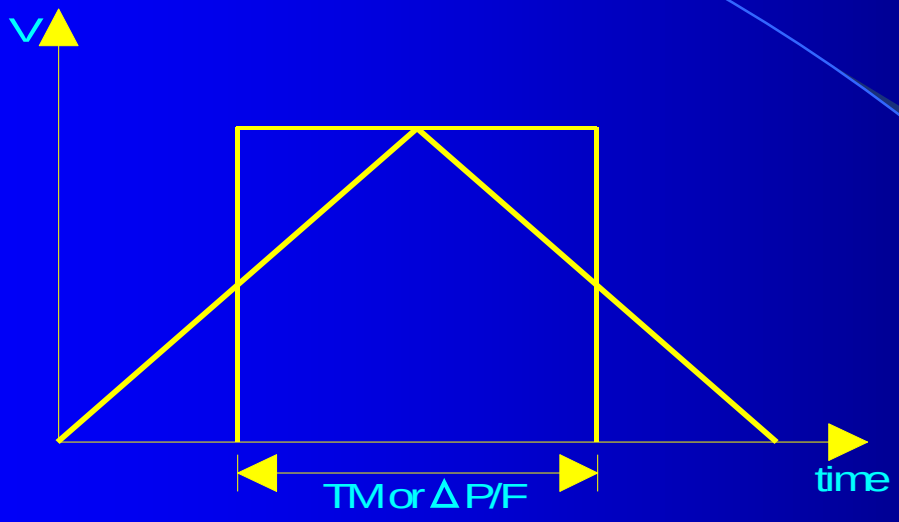
Linear 模式的轨迹(continued)
小Acceleration 时间

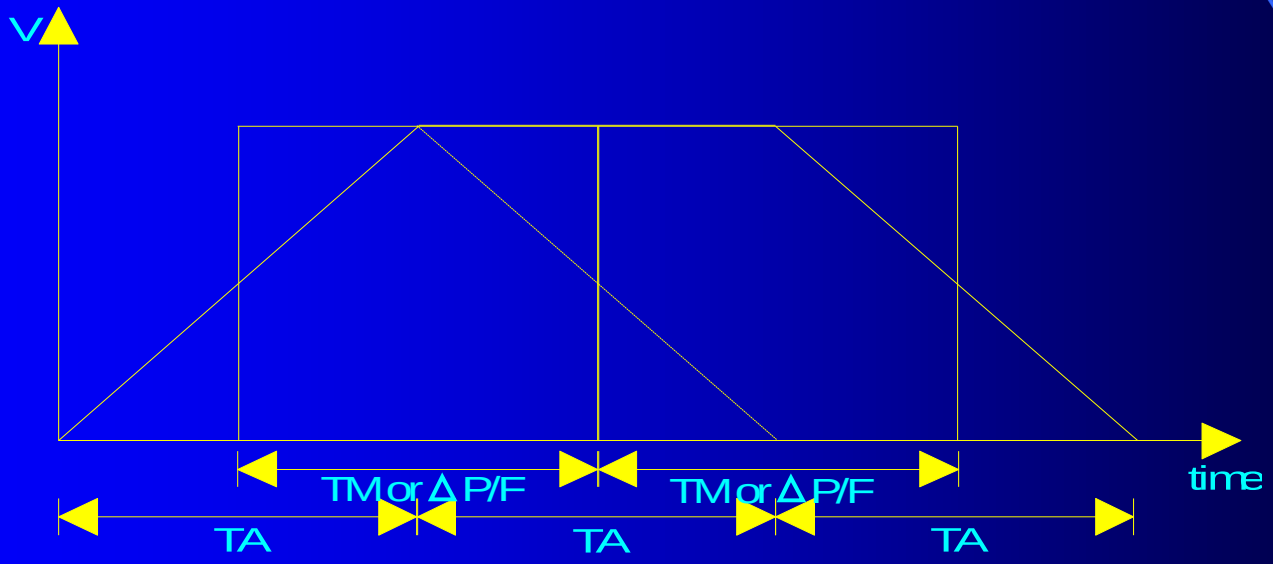




Linear 模式的轨迹(continued)

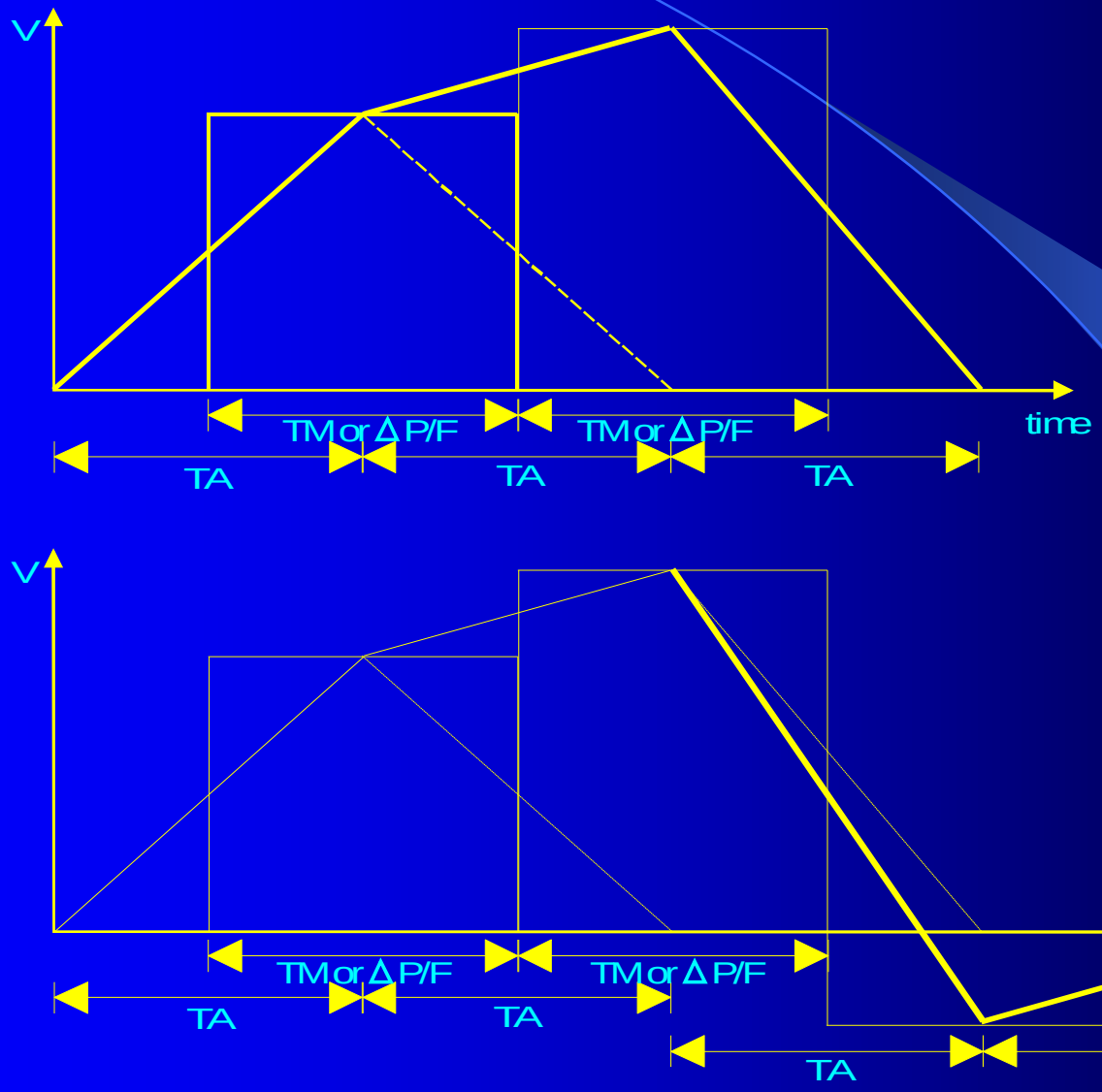
加速时间等于运动时间



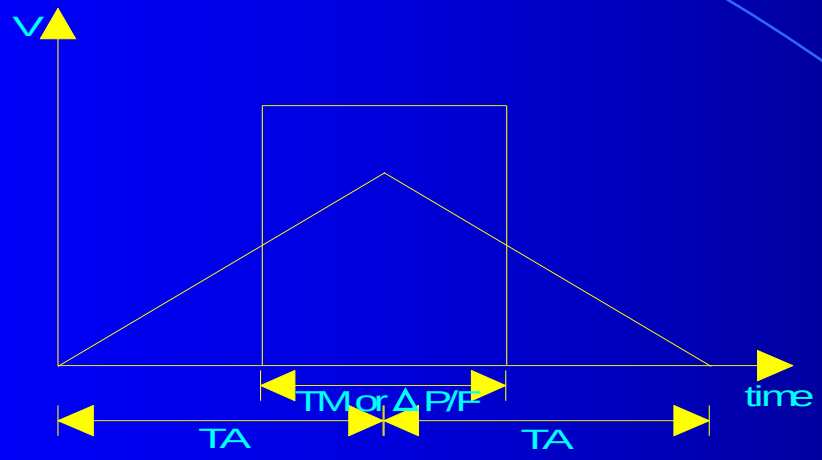


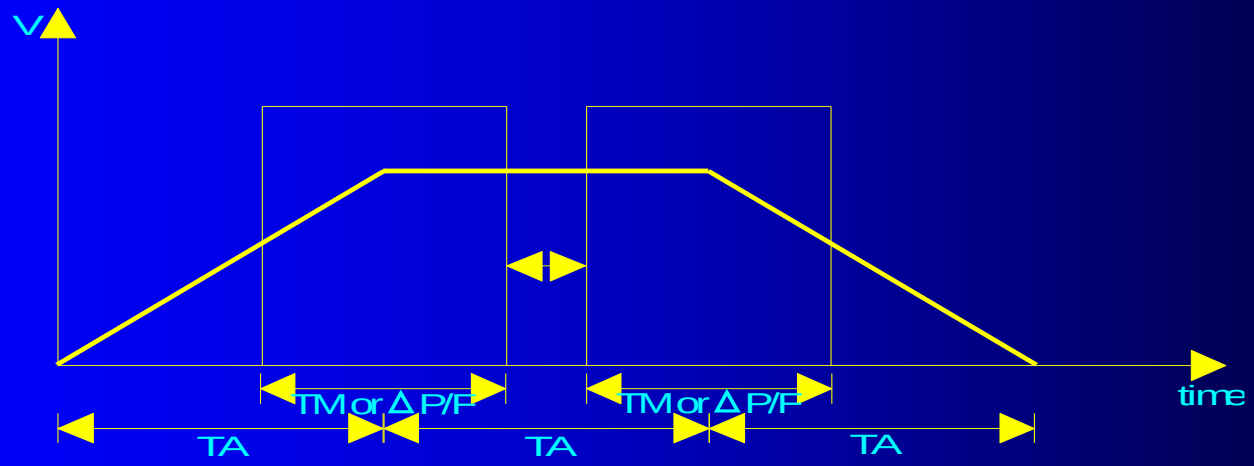
Linear 模式的轨迹(continued)

加速时间等于运动时间

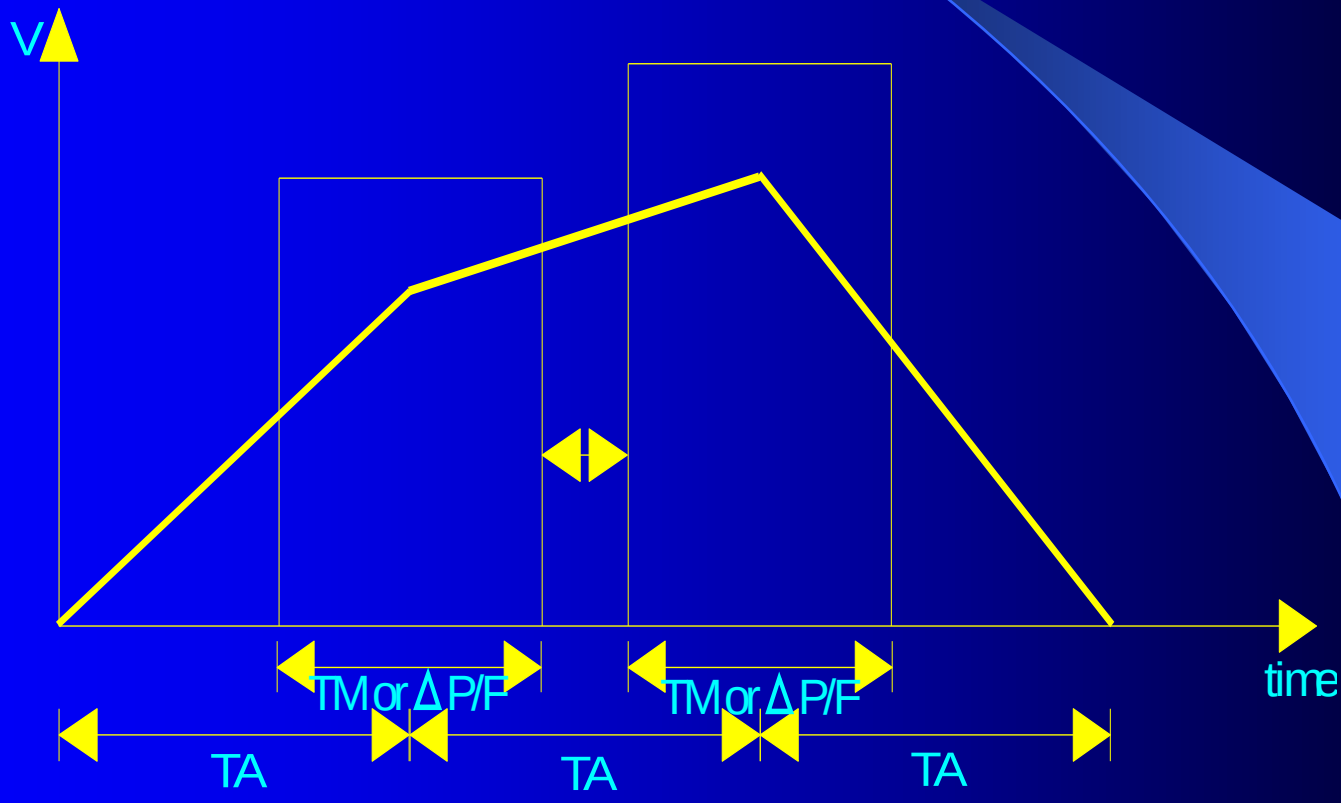


Linear 模式的轨迹
大 (速度限制) 加速时间





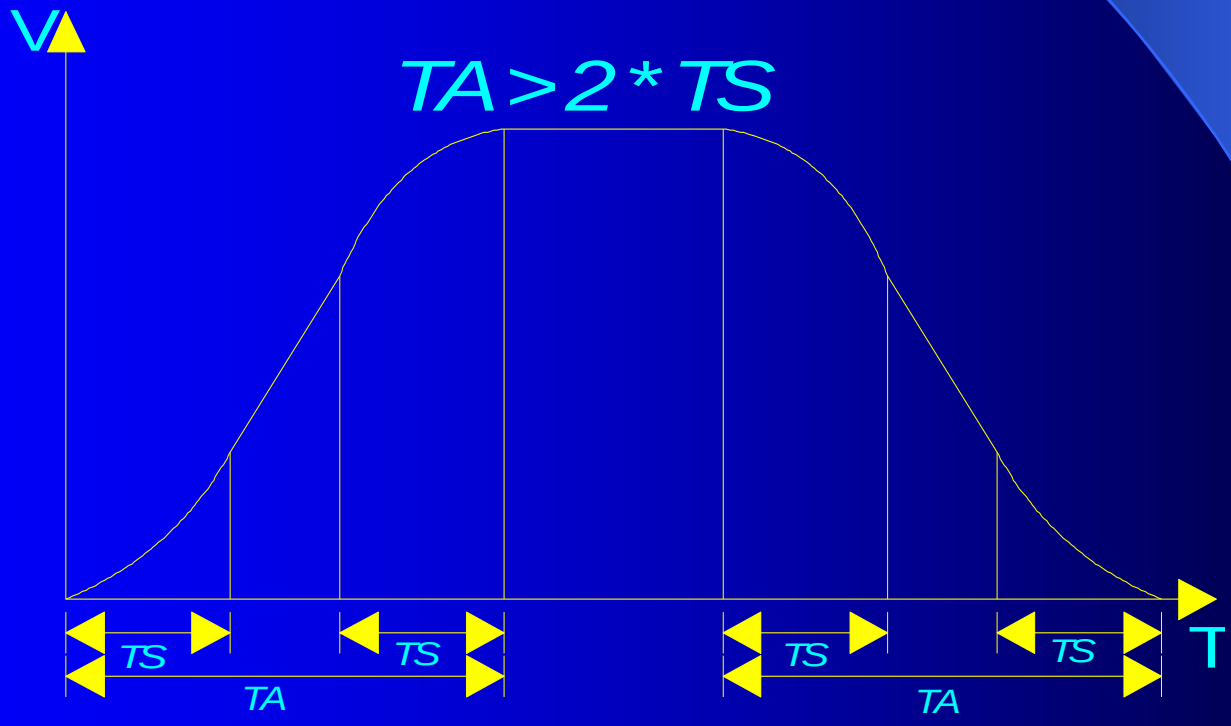
Linear 模式的轨迹
大 (速度限制) 加速时间



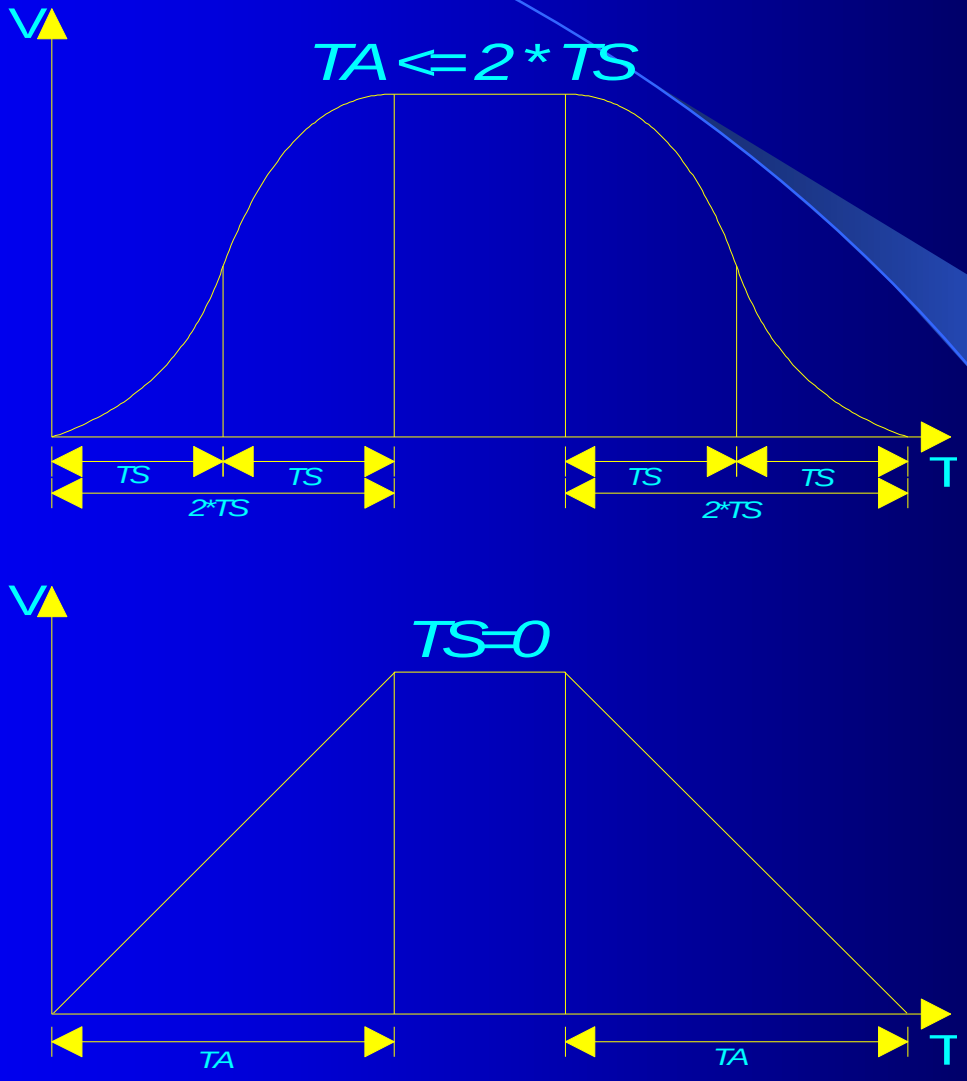
加速参数

TA 编程加速时间
(单位: msec); 整数

TS 编程S-曲线时间
(单位: msec); 整数



加速参数(continued)



例 3: 一个复杂的运动

This example introduces incremental and time-specification of moves, looping logic, using variables, scaling of axes, and simple arithmetic. Note that logical and mathematical operations do not delay moves.

```
;******************** Set-up and Definitions ********************

&1          ; Coordinate system 1
CLOSE       ; Make sure all buffers are closed
#1->1000X ; 1 unit (cm) of X is 1000 counts of motor 1

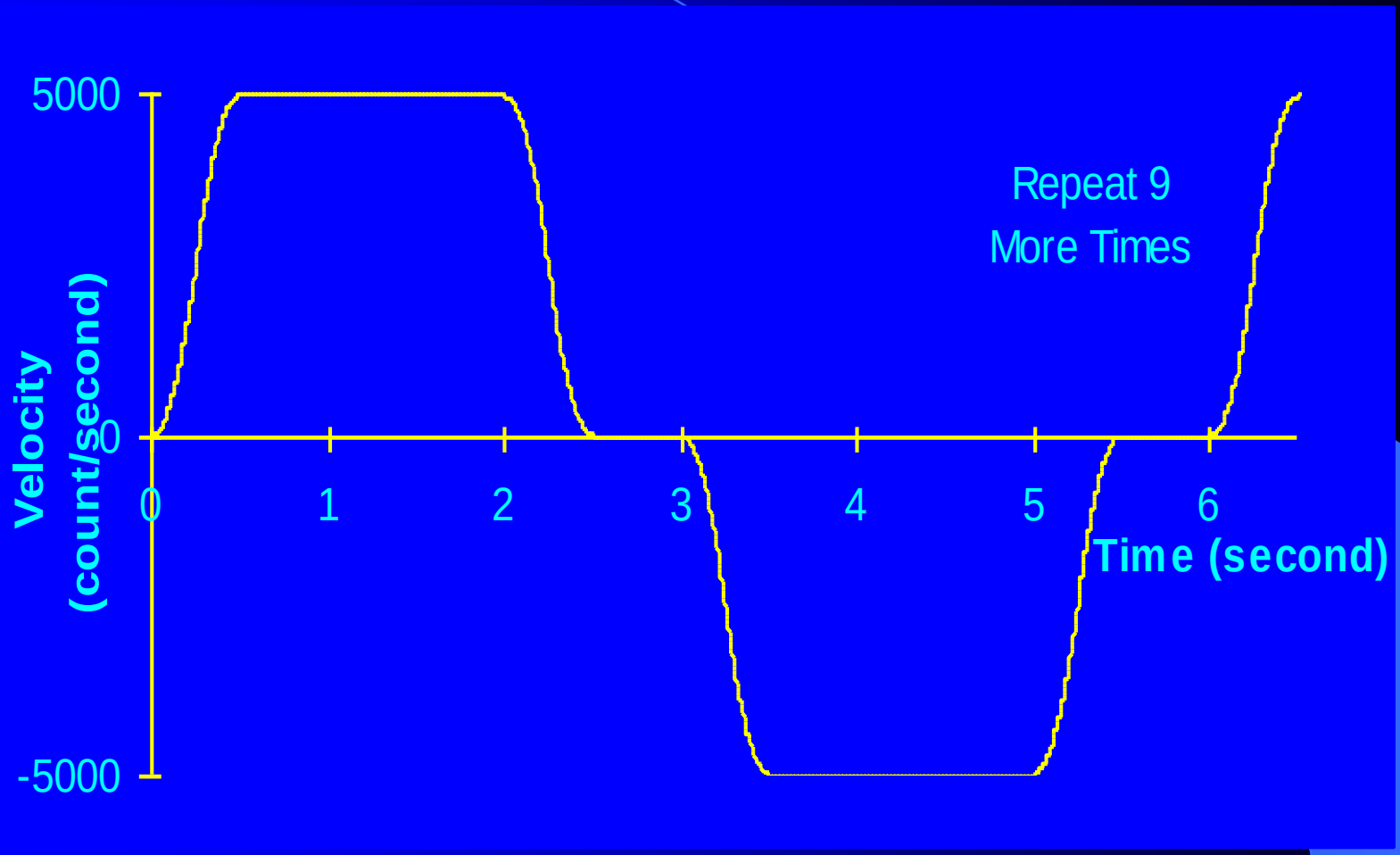
;******************** Motion Program Text ************************

OPEN PROG 2           ; Open buffer for entry, Program #2
CLEAR       ; Erase existing contents of buffer
LINEAR      ; Blended linear interpolation move mode
INC         ; Incremental mode - moves specified by distance
TA500       ; 1/2 sec (500 msec) acceleration time
TS250       ; 1/4 sec in each half of S-curve
TM2000      ; 2 sec move time (to start of decel)
P1=0        ; Initialize a loop counter variable
WHILE (P1<10)         ; Loop until condition is false (10 times)
  X10       ; Move X-axis 10 cm (=10,000 cts) positive
  DWELL500            ; Hold position for 1/2 sec
  X-10      ; Move X-axis back 10 cm negative
  DWELL500            ; Hold position for 1/2 sec
  P1=P1+1 ; Increment loop counter
ENDWHILE    ; End of loop
CLOSE       ; Close buffer - end of program
```

运行程序:

```
&1 B2 R    ; 坐标系1, 指向运动程序2,运行
```

例 3: 一个复杂的运动


5000
0
-5000
Velocity
(count/second)
0
1
2
3
4
5
6
Time (second)
Repeat 9
More Times

直线插补测验

曲线的水平和垂直轴的含义?

什么是 TS, TA, TM and F?

TM 和 F 的关系是什么?

以100msTA时间和20ms的TS时间，TM为500ms的时间外成一个运动，总的运动时间是多少?

如果程序表达 TS100 TA100 TM100，完成3个混合运动的总时间是多少?

PMAC 运动速度混合

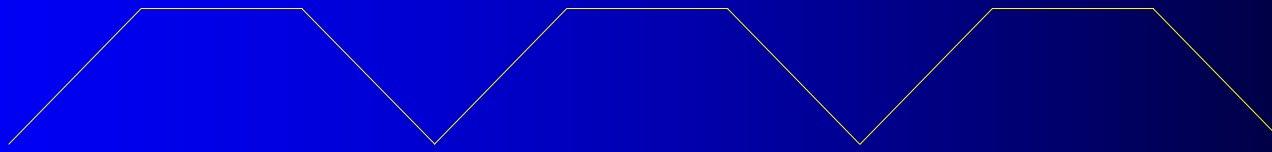
PMAC 在如下情况下速度无法混合:

- 两个运动指令中间有 **DWELL指令**
- 2 个向后跳转指令 (GOTO, ENDW)
- 速度混合功能无效设定 **(Ix92=1)**

Blending allowed - Ix92 = 0, no DWELL's, no double jumpback



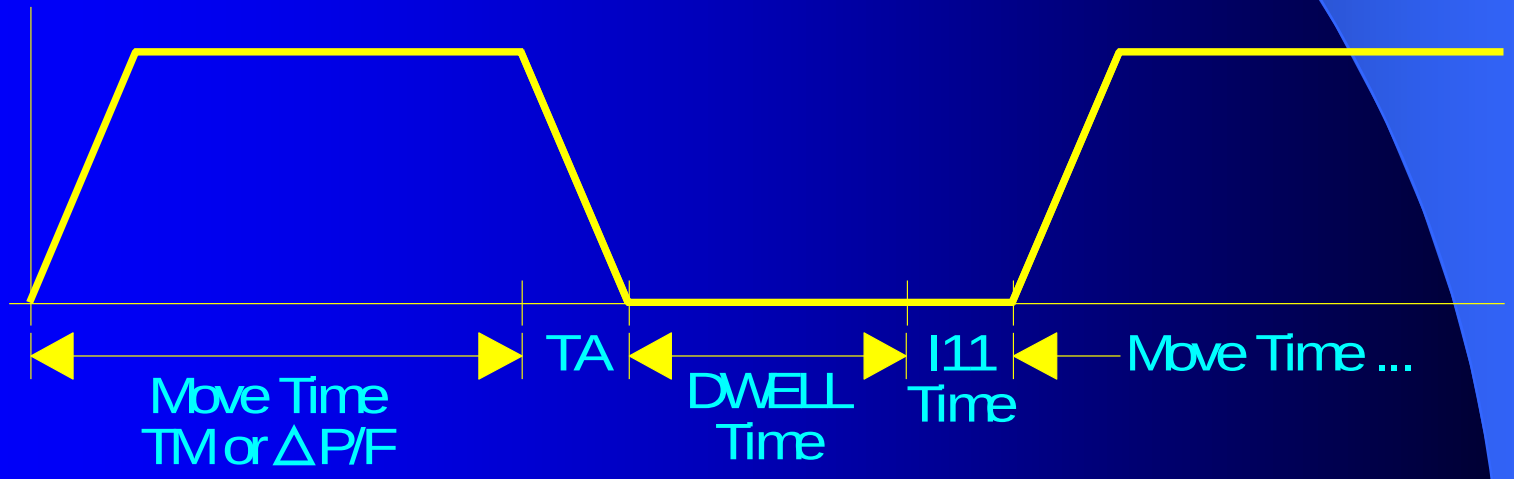
Blending not allowed - Ix92 = 1, DWELL's or double jumpback



DWELL Vs. DELAY

DWELL

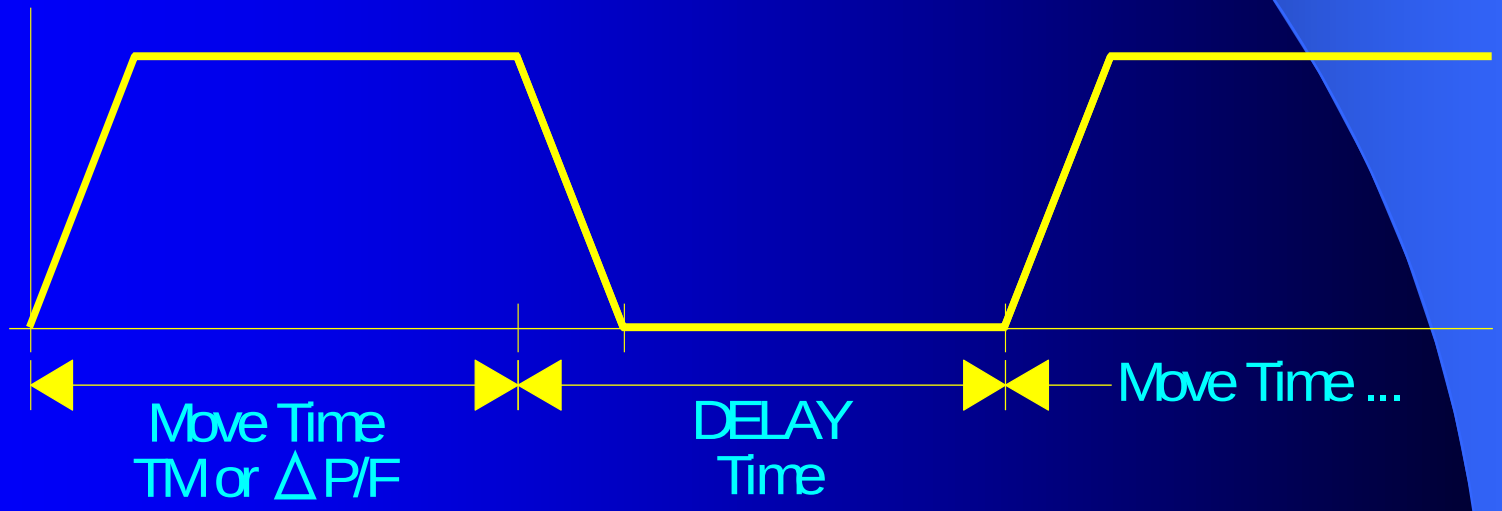
总使用固定的时基 (I10)
暂停时间不包括减速过程
直到DWELL结束才执行下面运动的计算 (add I11 time)



DWELL Vs. DELAY (continued)

DELAY

使用可变的时基 (% value)
暂停时间包括减速过程
最小暂停时间是当前 TA时间
下面运动时间开始于DELAY指令



例4：速度混合有效的例程

This example shows how to program a blended move on the PMAC and the function of Ix92. First the program specifies how to do the move, then commands the move.

```
****************** Set-up and Definitions ******************

DEL GAT    ; Erase any defined gather buffer
&1         ; Coordinate System 1
CLOSE      ; Make sure all buffers are closed
#1->X      ; Assign motor 1 to the X-axis - 1 program unit
           ; of X is 1 encoder count of motor #1
I192=0     ; Enable blending

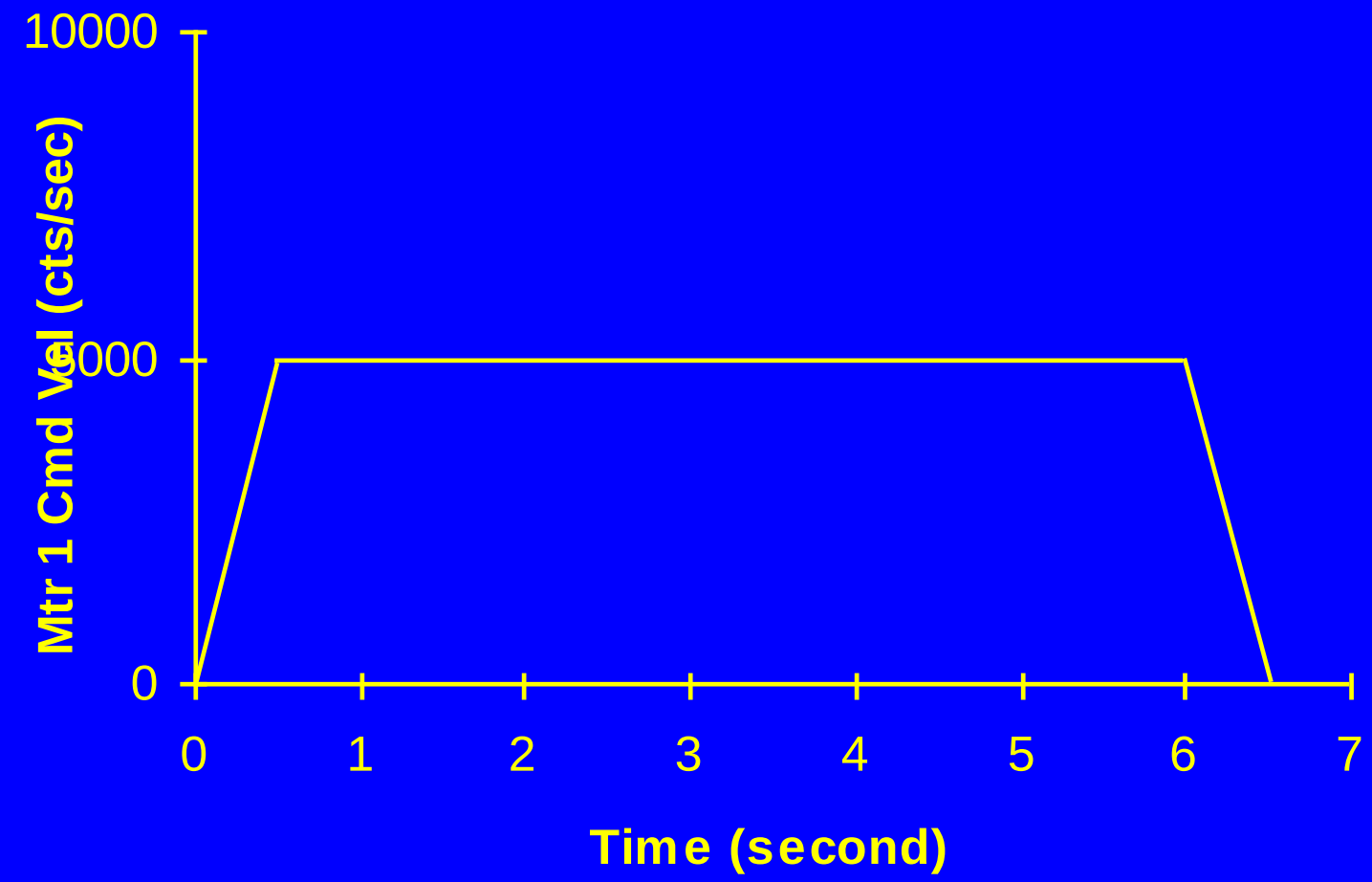
****************** Motion Program Text **********************

OPEN PROG 1          ; Open buffer for program entry, Program #1
CLEAR      ; Erase existing contents of buffer
LINEAR     ; Blended linear interpolation move mode
ABS        ; Absolute mode - moves specified by position
TA500      ; Set 1/2 sec (500 msec) acceleration time
TS0        ; Set no S-curve acceleration time
F5000      ; Set feedrate (speed) of 5000 units(cts)/sec
X10000     ; Move X-axis to position 10000
X20000     ; Move X-axis to position 20000
X30000     ; Move X-axis to position 30000
DWELL0     ; Stop program lookahead
CMD"ENDG"  ; Send On-line command to stop data gathering
CLOSE      ; Close buffer - end of program
```

To run this program:

```
DEF GAT<CR>
GAT &1 B1 R          ; GATHER, CS. 1, point to Beginning of Program 1, Run
```

例4：速度混合有效的例程



例5：速度混合无效的例程

This example shows how to program moves on the PMAC and the function of Ix92. First the program specifies how to do the move, then commands the move.

```
****************** Set-up and Definitions ******************

DEL GAT    ; Erase any defined gather buffer
&1         ; Coordinate System 1
CLOSE      ; Make sure all buffers are closed
#1->X      ; Assign motor 1 to the X-axis - 1 program unit
           ; of X is 1 encoder count of motor #1
I192=1     ; Disable blending

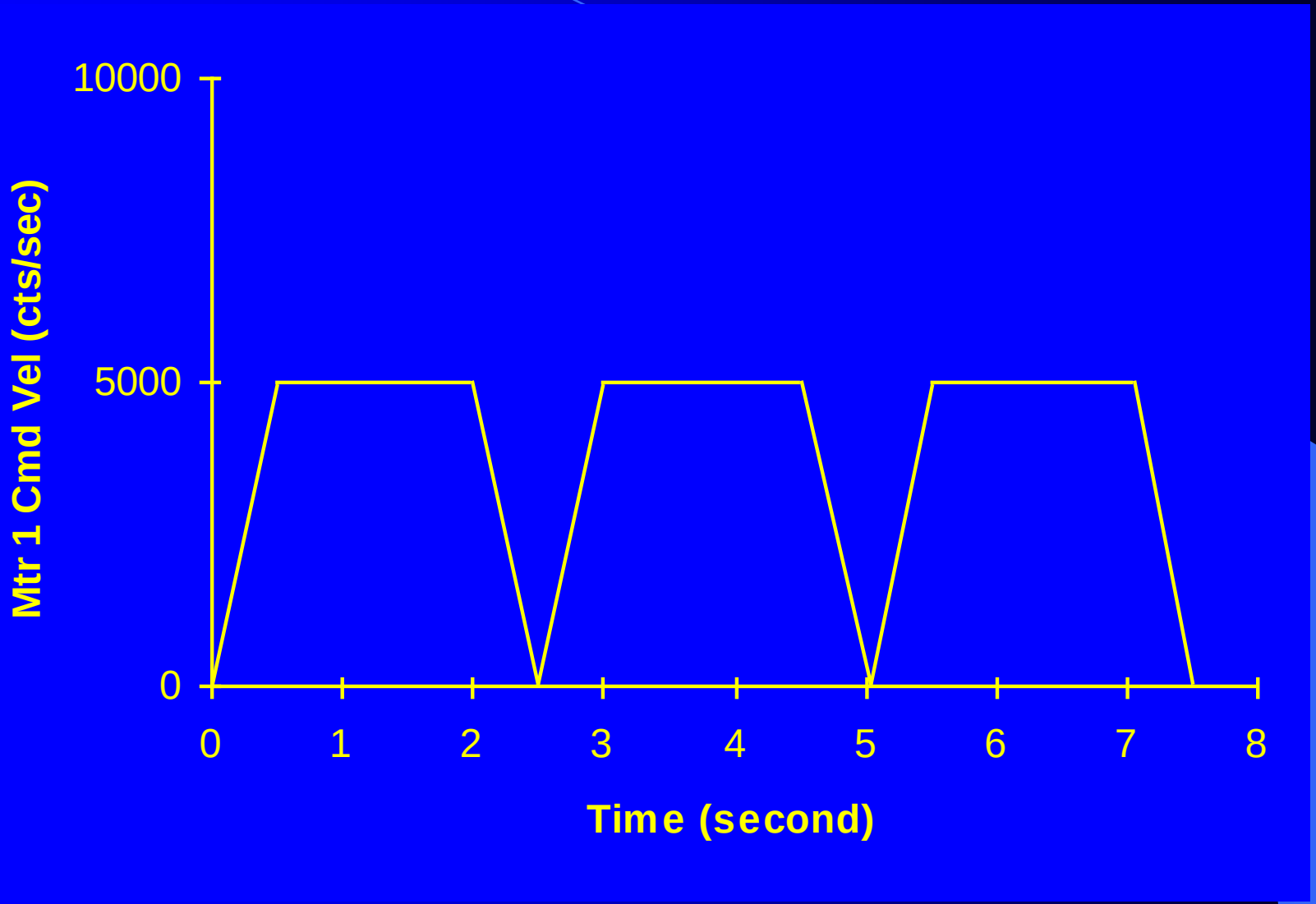
****************** Motion Program Text **********************

OPEN PROG 1          ; Open buffer for program entry, Program #1
CLEAR      ; Erase existing contents of buffer
LINEAR     ; Blended linear interpolation move mode
ABS        ; Absolute mode - moves specified by position
TA500      ; Set 1/2 sec (500 msec) acceleration time
TS0        ; Set no S-curve acceleration time
F5000      ; Set feedrate (speed) of 5000 units(cts)/sec
X10000     ; Move X-axis to position 10000
X20000     ; Move X-axis to position 20000
X30000     ; Move X-axis to position 30000
DWELL0     ; Stop program lookahead
CMD"ENDG"  ; Send On-line command to stop data gathering
CLOSE      ; Close buffer - end of program
```

To run this program:

```
DEF GAT<CR>
GAT &1 B1 R          ; GATHER, CS. 1, point to Beginning of Program 1, Run
```

例5: 速度混合无效的例程



速度混合测验

使用 速度指令 (**F**) 在增量方式 (**INC**)， 指令 **X1000** 和指令 **X100 X400 X500**如果 Ix92=0, TA< TM or (TA<ΔP/F), 和 2×TS<TA有何区别?

矢量速率

PMAC 可以自动计算运动时间（TM)使电机运行与希望的速度速率下.

这用于多个电机在一个坐标系下控制一个刀具，来自动分配各自的速度

需要学习:

速率指令的表达

PMAC如何计算速率

矢量速度轴

```
INC
FRAX (X,Y)
X3 Y4 F10
```

$$\text{Dist} = \sqrt{3^2 + 4^2} = 5$$

$$\text{Move Time} = 5 / 10 = 0.5$$

$$V_x = 3 / 0.5 = 6$$

$$V_y = 4 / 0.5 = 8$$

```
INC
FRAX (X,Y)
X3 Y4 Z12 F10
```

$$\text{Dist} = \sqrt{3^2 + 4^2} = 5$$

$$\text{Move Time} = 5 / 10 = 0.5$$

$$V_x = 3 / 0.5 = 6$$

$$V_y = 4 / 0.5 = 8$$

$$V_z = 12 / 0.5 = 24$$

矢量速度轴(续)

```
INC
FRAX (X,Y,Z)
X3 Y4 Z12 F10
```

$$\text{Dist} = \sqrt{3^2 + 4^2 + 12^2} = 13$$

$$\text{Move Time} = 13 / 10 = 1.3$$

$$V_x = 3 / 1.3 = 2.31$$

$$V_y = 4 / 1.3 = 3.08$$

$$V_z = 12 / 1.3 = 9.23$$

```
INC
FRAX (X,Y)
C10 F10
```

$$\text{Dist} = 0$$

$$\text{Move Time} = 0 / 10 = 0 \ (< \text{TA})$$

Acceleration - limited move

速度时间单位

Ix90　速度时间单位
(单位: msec)

$$\text{F UNIT} = \frac{\text{LENGTH UNIT}}{\text{TIME UNIT}} \Rightarrow \frac{\text{AXIS DEF. SCALING}}{\text{Ix90 (msec)}}$$

例:　`#1 -> 2000x`
`Ix90 = 1000`

$$\text{F UNIT} = \frac{\text{2000 cts}}{\text{1000 msec}} \Rightarrow \frac{\text{mm}}{\text{sec}}$$

速率测验

如果坐标系1 的 I190=60000，那么速率时间单位是什么?

如果X 和 Y 是速率轴, 如果指令 **X100 Y200**，各自速度如何分配?

如果你编程指令 **F5** 定义速度为 5 inches/sec,如何设定 Ix90和轴的定义?

Exercise 1: Indexing a Motor

Problem:

You have been asked to use PMAC in an indexing application, PMAC must index motor #1, which drives a leadscrew, a distance of 10 cm, pause at this location for 4 seconds then return to its original location. Each of the moves must take place in 2 seconds.

The program will be started by an ASCII command string sent to PMAC from a Programmable Logic Controller over the RS232 port.

Hardware:

2000 counts per revolution encoder

5:1 gear reduction

1cm/rev pitch leadscrew

SET Ix92=0 to enable blending

Exercise 2: Indexing a Motor

Problem:

As is the case with all applications, the requirements have been changed. The following must be added to the basic requirements of exercise 1.

The desired motion profile has been changed. The system must now move from its zero location to 10 cm, pause for 4 seconds and then move to 20 cm in 2 more seconds and pause for 1 second. It then must return to 0 cm in 1.5 seconds.

The system has now become a multi-axis project. Not one but four identical leadscrews must simultaneously execute this profile.

Exercise 2: Indexing a Motor (continued)

To minimize wear and tear on the mechanical components, instantaneous changes in acceleration (jerk) must be eliminated.

This machine will be sold to a European company which will write some new programs for it. They only want to program in centimeters so your programs must also use centimeters.

A feedrate override switch has also been added to the system. Your programmed pauses must now change as the feedrate override changes. This means that a 1.5 second pause will last 3 seconds at 50% feedrate override. Use `M197->X:$0806,0,24,s` for feedrate override control.

Exercise 3: Conditional Branching

Problem:

The European Company is having a lot of success with your machine and your boss now has some extra R&D money. He wants you to write another application based on the same platform.
2 input switches have been connected to PMAC's machine inputs 1 and 2. He wants these switches to control the motion of the stages as follows.

If machine input 1 is on, the stages should move from 0 cm to 10 cm at a speed of 1 cm/sec, pause for 1 second then return to 0 cm at the same speed and pause for 100 milliseconds until looking for the next input.

Exercise 3: Conditional Branching (continued)

If machine input 2 is on, the stages should move from 0 cm to -10 cm at a speed of 2 cm/sec, pause for 1 second then return to 0 cm at the same speed and pause for 100 milliseconds until looking for the next input.

If both machine inputs 1 and 2 are on, or if neither are on, the stages should not move from 0 cm and should pause for 100 milliseconds until looking for the next input.

圆弧插补

PMAC 允许 X,Y, 和 Z 轴在一个坐标系下执行圆弧插补指令。
TS 可控制起始和结束的加减速时间. 圆弧插补的混合速度由指令(F) 或时间 TM来定义.

需要学习:

- 什么变量控制圆弧插补
- 如何使用圆弧插补
- 如何定义插补平面

圆弧插补

圆弧插补的通常定义:

1) 运动分段时间 — Typical value: I13 = 10

2) 插补平面定义 — NORMAL {vector} {data} …

3) 终点坐标定义方式 — ABS / INC

(4) 圆心矢量的定义方式 — ABS (R) / INC (R)

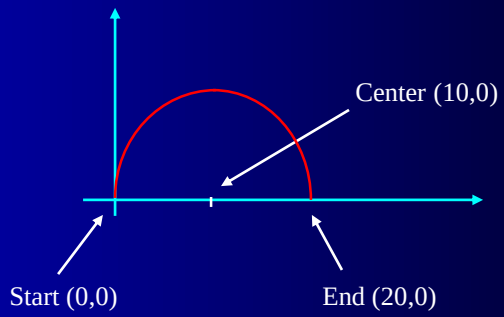
(if vector mode definition is used)

(5) 圆弧方向定义 — CIRCLE 1 / CIRCLE 2

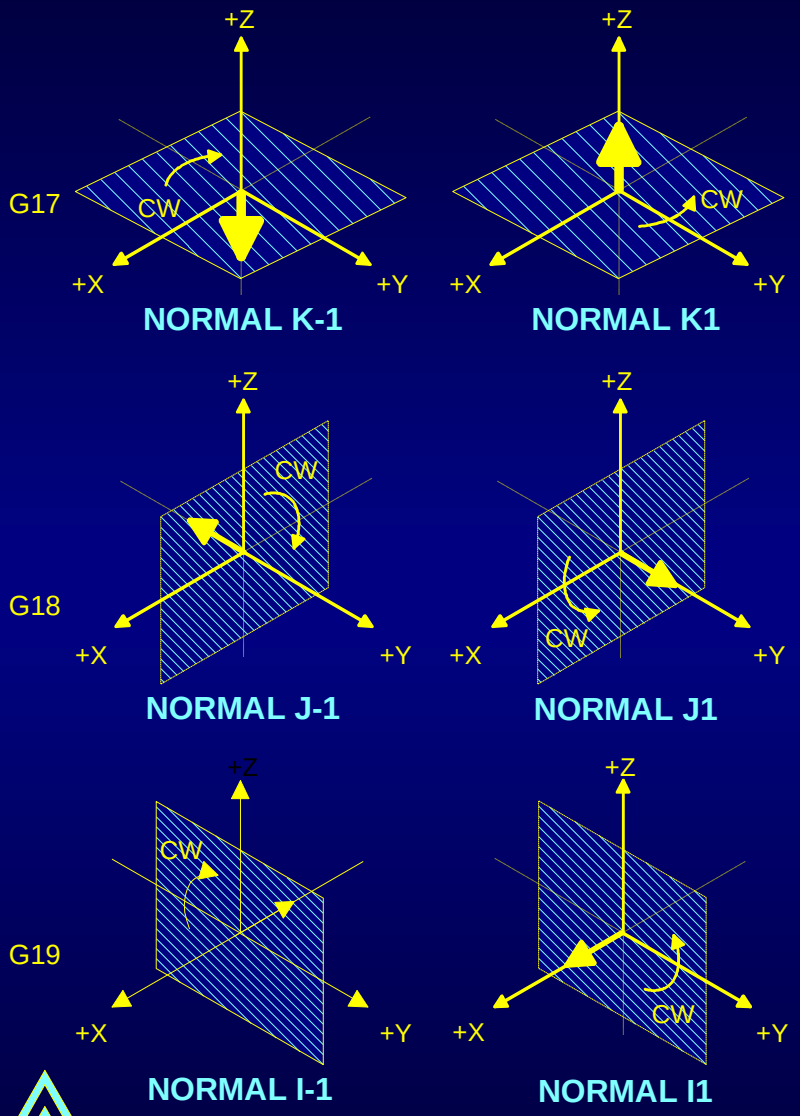
(6) 圆弧指令 — X{Data} Y{Data} I{Data} J{Data} / X{Data} Y{Data} R{Data}

例:

```
I13=10              ;Move Segmentation Time
NORMAL K-1          ;XY plane
INC                 ;Incremental End Point definition
INC(R)              ;Incremental Center Vector definition
CIRCLE 1            ;Clockwise circle
X20 Y0 I10 J0       ;Arc move
```



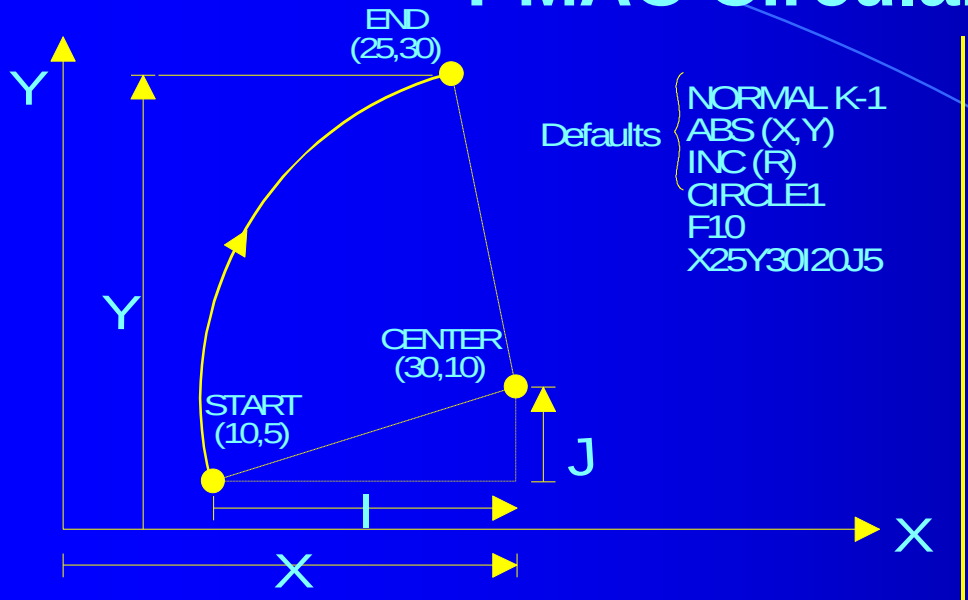
CIRCULAR INTERPOLATION



NORMAL VECTORS FOR CIRCULAR MOVES:
THE PLANES AND CLOCK WISE ARCS THEY DEFINE

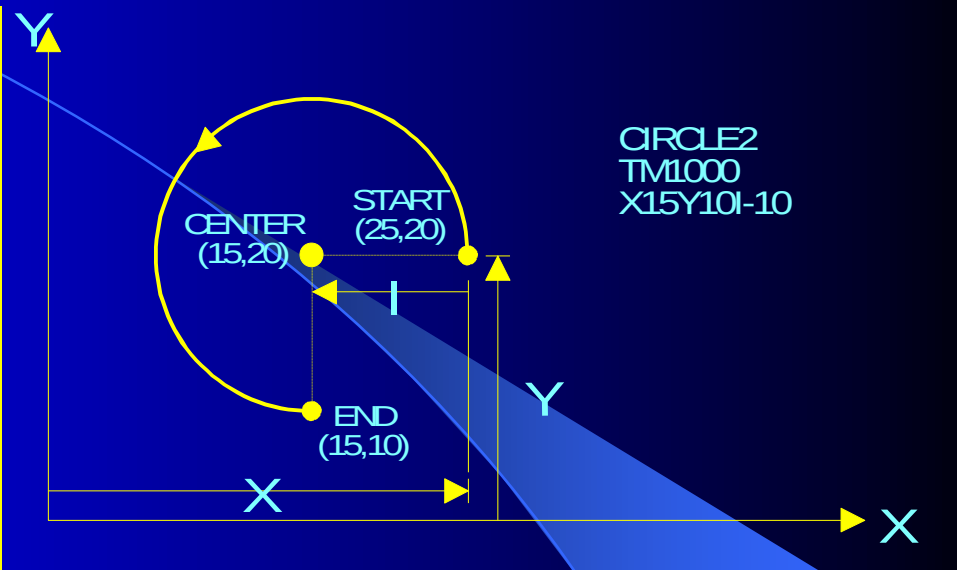
(NRM-VEC)

PMAC Circular Interpolation

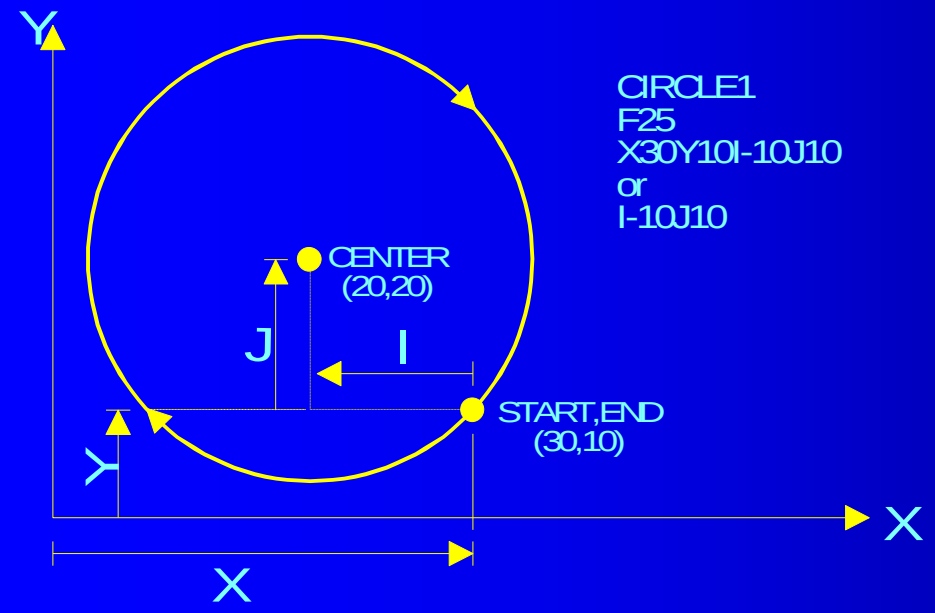


Defaults
- NORMAL K-1
- ABS (X,Y)
- INC (R)

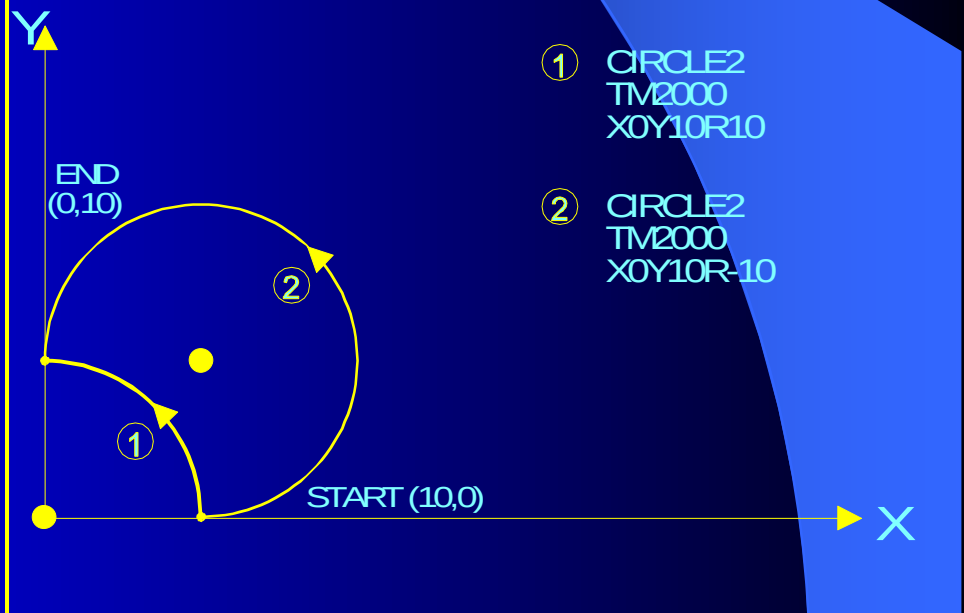
CIRCLE1
F10
X25Y30I20J5



CIRCLE2
TM1000
X15Y10I-10



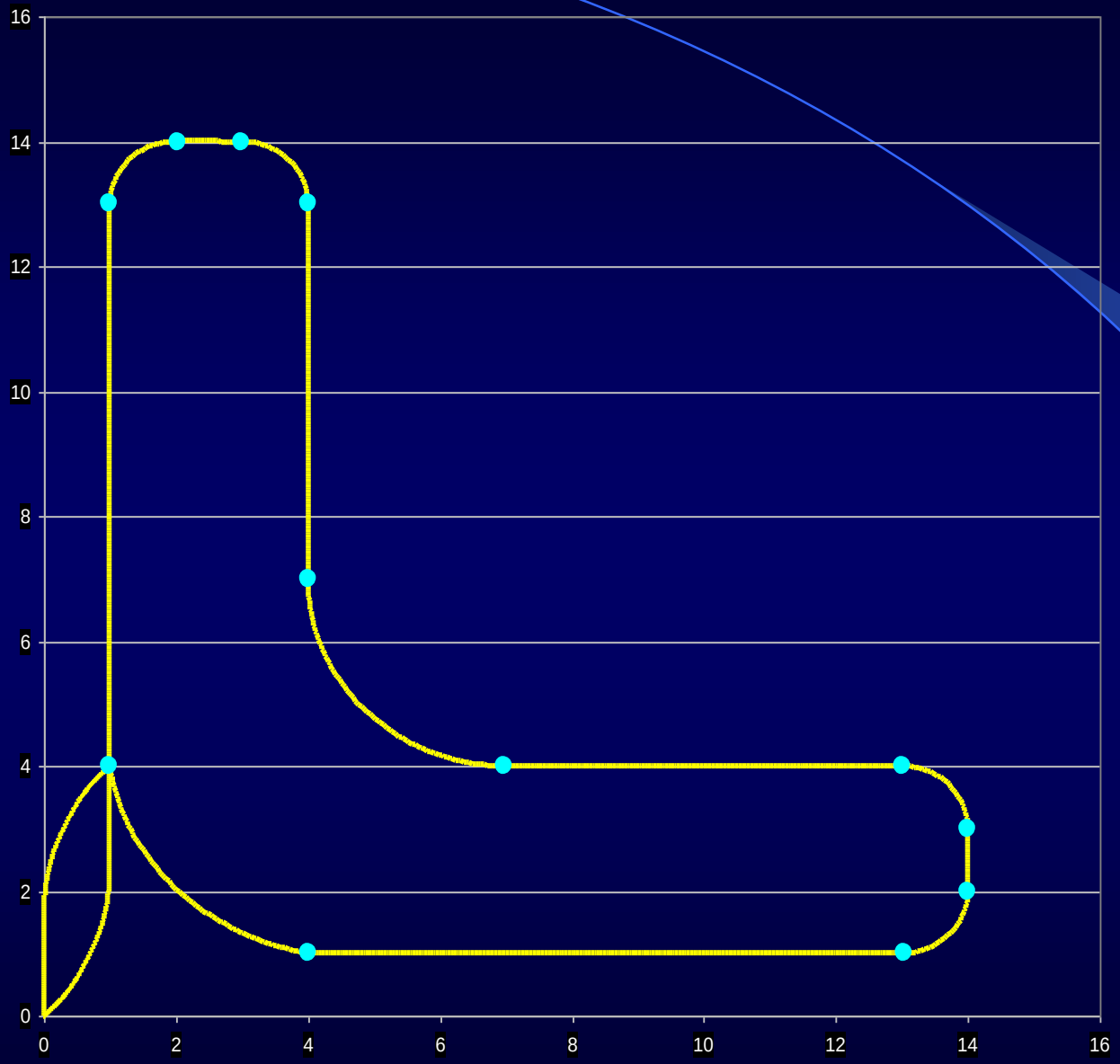
CIRCLE1
F25
X30Y10I-10J10
or
I-10J10



① CIRCLE2
TM2000
X0Y10R10

② CIRCLE2
TM2000
X0Y10R-10

圆弧插补例程




```
;Setup and definitions
&1
#3->10000x
#4->10000y
i13=10

;Motion Program Text

open prog4 clear
normal K-1
rapid x1 y4
f5
linear y13
circle1 x2 y14 i1 j0
linear x3
circle1 x4 y13 i0 j-1
linear y7
circle2 x7 y4 i3 j0
linear x13
circle1 x14 y3 i0 j-1
linear y2
circle1 x13 y1 i-1 j0
linear x4
circle1 x1 y4 i0 j4
dwell100
rapid x0 y0
close
```

提前运算（前瞻功能）
PRECALCULATION

在 blend 和 spline 运动中, 运动程序必须将当前运动后几步的运动指令提前进行运算.

根据不同的运动模式，在程序运行中将进行0，1，或2段运动指令的前瞻计算.

我们将要学习:

- 前瞻功能做了些什么？
- 怎样停止前瞻功能？
- 如何使用同步变量？

PRECALCULATION

All program calculations and assignments, except for synchronous M-variable assignments, between the move in progress and the move being calculated are performed one line at a time during the look-ahead.

This may be a problem with M-variables, particularly outputs, as the action will take place sooner than expected.

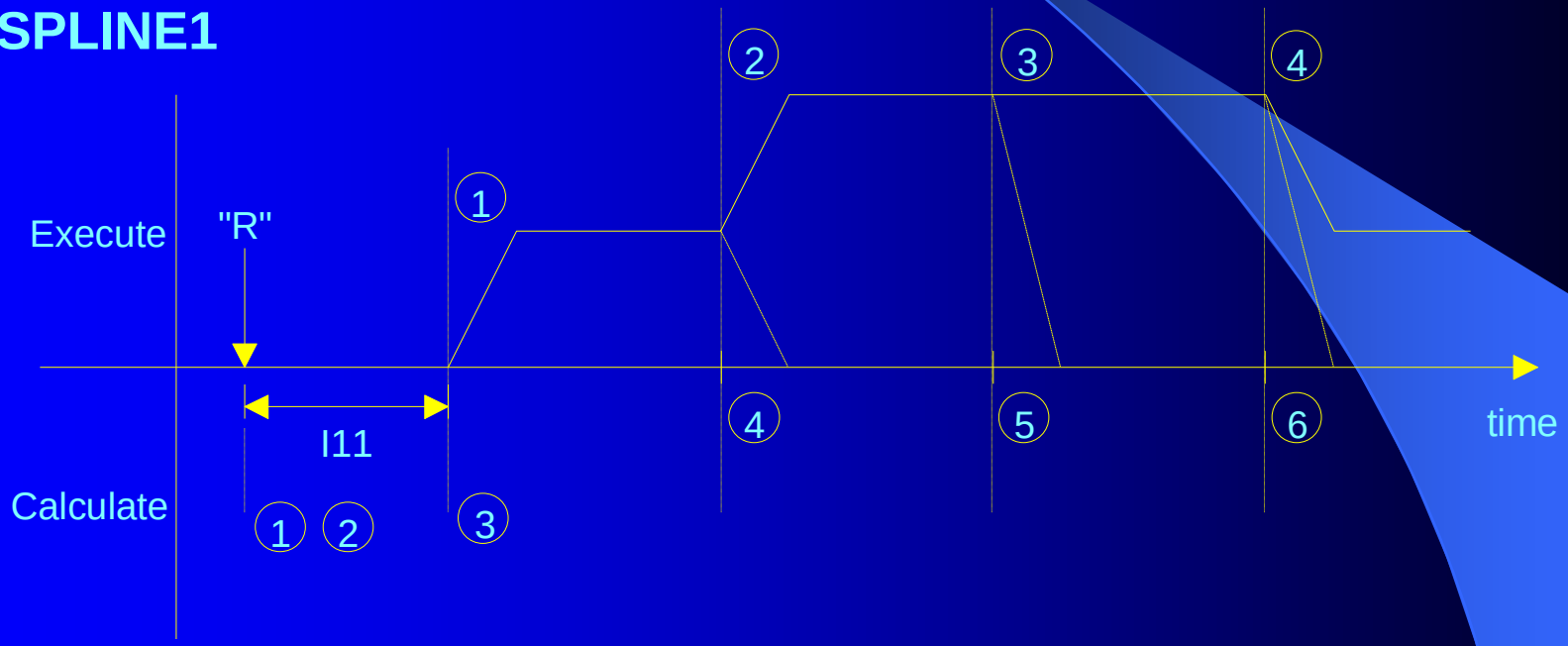
Example:

```
LINEAR      ;linear move mode
X10         ;move X-axis to 10
X20         ;move X-axis to 20
M1=1        ;turn on output #1
X50         ;move X-axis to 50
```

The output M1 will be turned on at the beginning of the X10 move due to PMAC's precalculation of the program

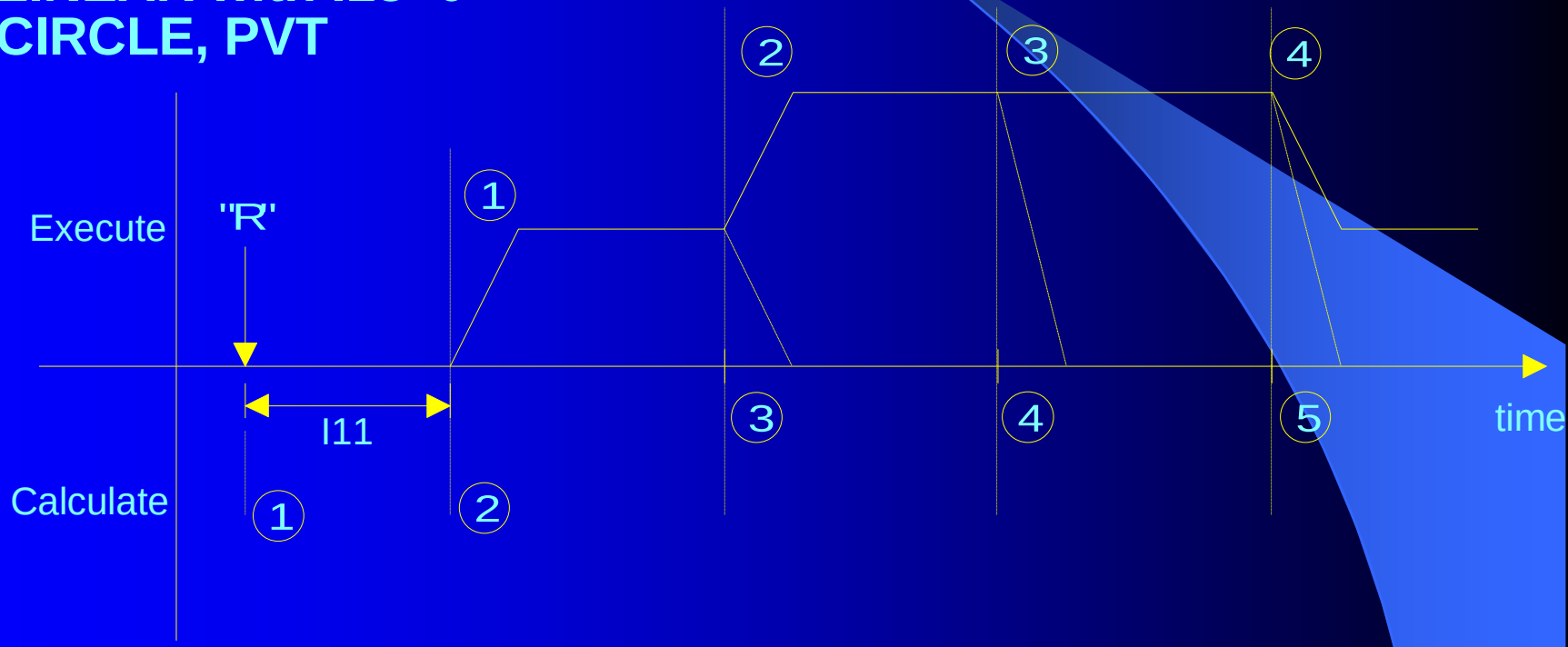
PMAC Motion Program Precalculation

A. Two moves ahead
LINEAR with I13=0
SPLINE1



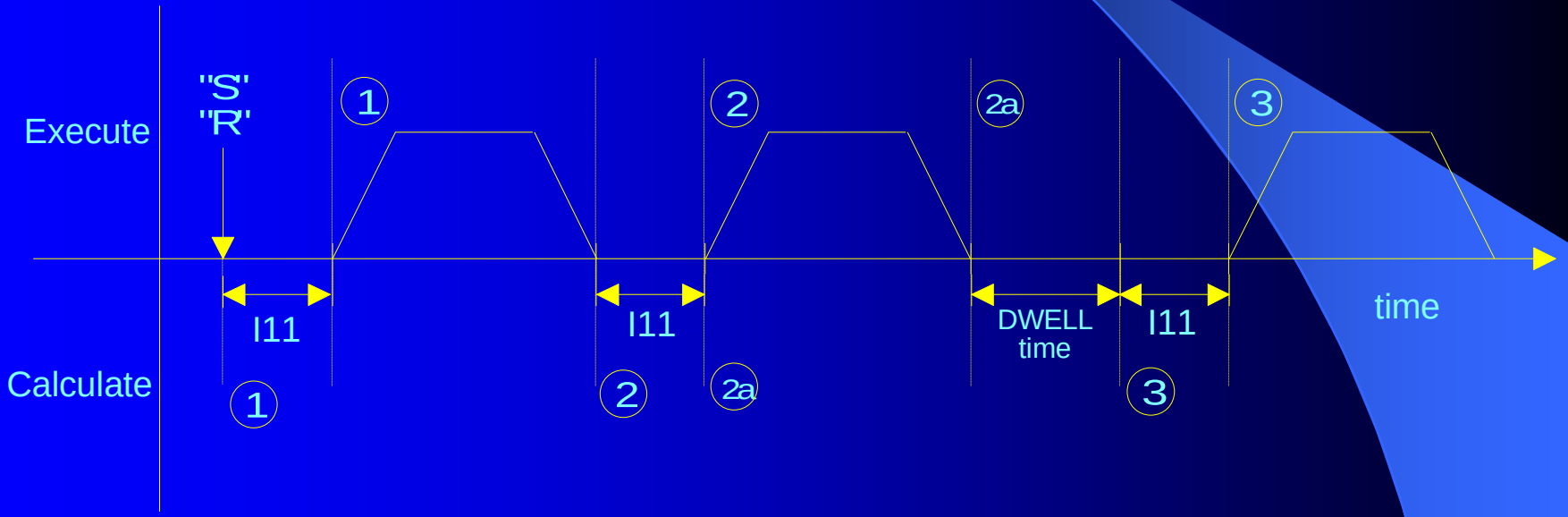
PMAC Motion Program Precalculation

B. One move ahead
LINEAR with I13>0
CIRCLE, PVT



PMAC Motion Program Precalculation

C. No moves ahead
Ix92=1, RAPID,
HOME, DWELL,"S"



同步 M-变量

Synchronous M-Variables assignments are a solution to the synchronization problems caused by move precalculation.

When a Synchronous M-variable assignment is encountered, it is not executed immediately; but put on a stack for execution at the start of the next move in the program. Output actions are thus synchronized with the motion action

Example:

```
LINEAR      ;linear move mode
X10         ;move X-axis to 10
X20         ;move X-axis to 20
M1==1       ;turn on output #1
X50         ;move X-axis to 50
```

The statement M1==1 is encountered at the beginning of the move to X10, but the action is not performed until the start of blending into the move X50

MACROS

Macros allow you to assign a specific name to a PMAC variable or command for programming purposes. For example, you can define the word PUMP_PRESSURE to the variable P10.

This allows you to use meaningful names in your programs which simplifies programming and memory management.

Macros are a feature of the Executive Program and are not a PMAC feature. When a file with Macros is downloaded to PMAC, the executive program substitutes the appropriate PMAC commands or variables in place of the Macros. PMAC only sees the PMAC commands and variables.

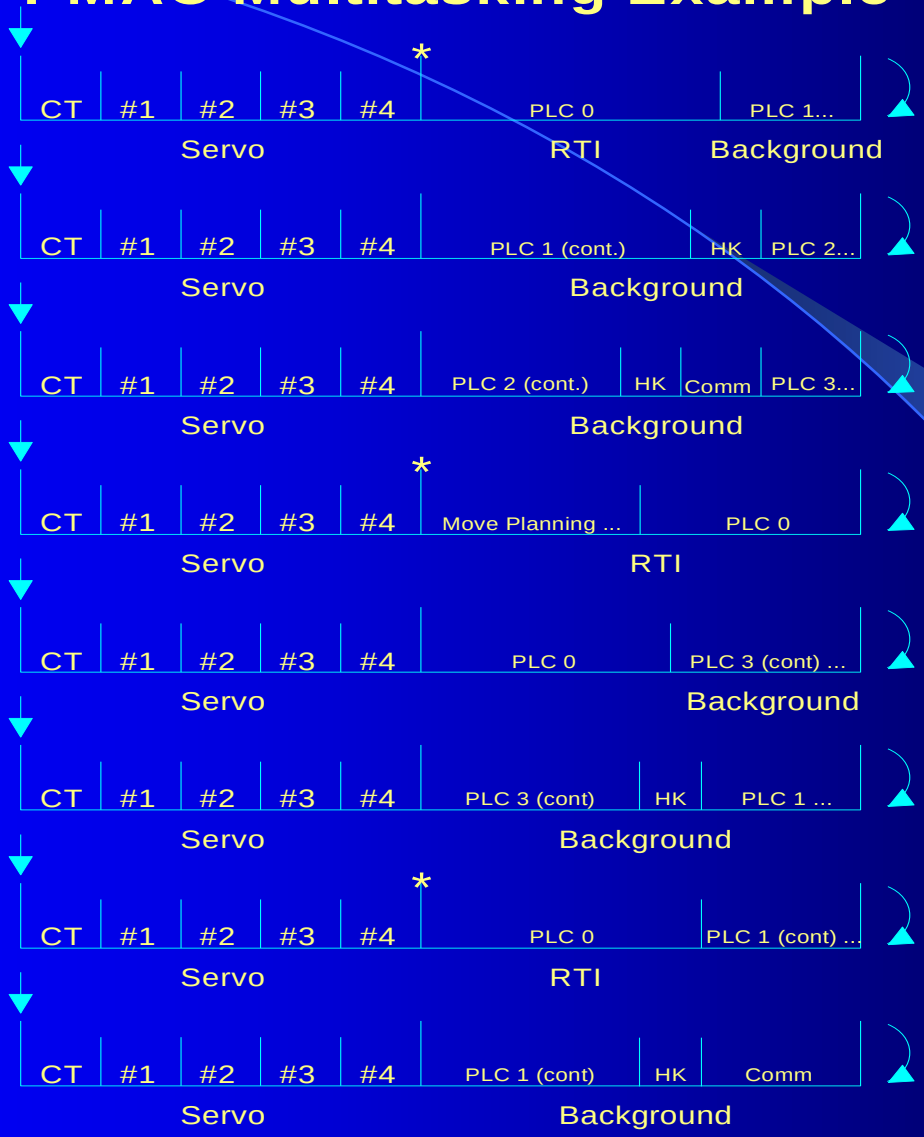
PMAC 计算优先级

1. **Single character I/O　单字符I/O**
   Serial, PC-bus, STD-bus; 200ns/char
2. **Commutation Update换相更新率**
   E29-E33 set (9KHz default); 3usec/axis
3. **Servo Update伺服更新率**
   E3-E6 set freq (2.25KHz default);30usec/axis
4. **Real Time Interrupt实时中断**
   I8 设定频率; 循环于:
   A. 运动程序的计算
      每次新运动的开始; 1 or 2 步提前
   B. PLC0
      当不执行运动程序
   C. PLCC0
   D. Dual Ported RAM双端口RAM
      Servo data buffers伺服数据存储器

PMAC 计算优先级

5. **VME mailbox处理器**
   事件驱动
6. 后台任务
   Done in time left over; rotates between:
   A. Communication line processing
      Event driven
   B. PLC1-31
      one at a time
   C. PLCC
      one scan through all PLCC's
   D. General Housekeeping
   E. Dual Ported RAM Data Buffers

PMAC Multitasking Example

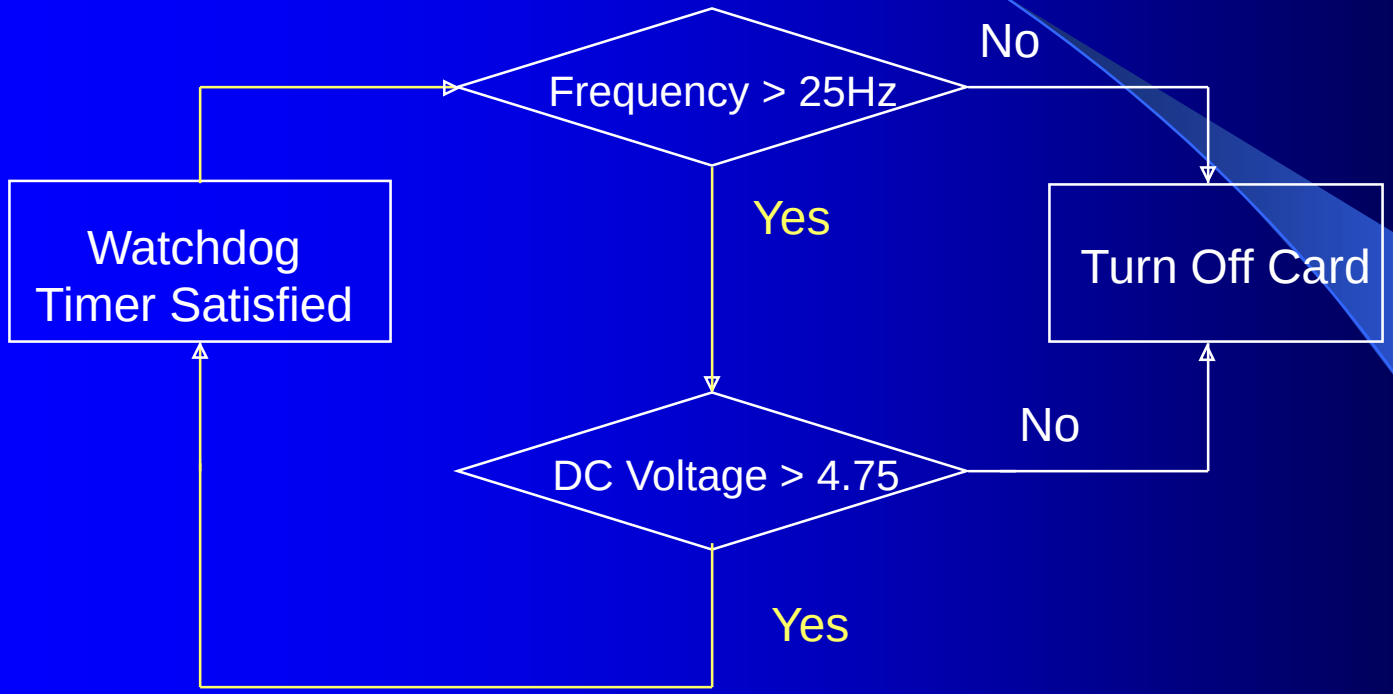


CT - Conversion Table
n - Motor n Servo Update
HK - Housekeeping

Comm - Communications Line Processing
▼ - Servo Interrupt
RTI - Real Time Interrupt Task
* - Start of RTI

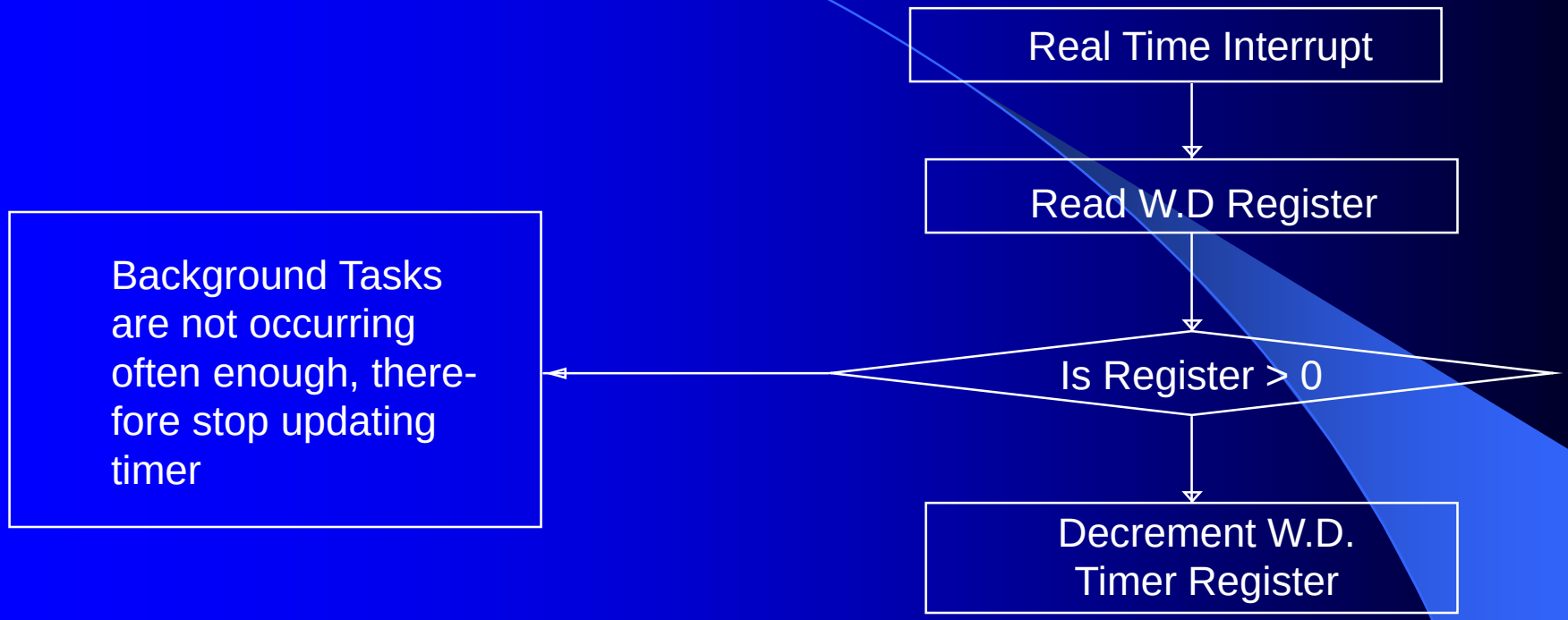
(mltitssk)

PMAC 看门狗计时器



PMAC algorithm to assure Watchdog gets 25 Hz signal and check for a minimum of 4.75V to the memory chips.

PMAC 看门狗计时器(续.)



PMAC algorithm to assure Watchdog gets 25 Hz signal. This ensures both background and foreground tasks are done reasonably often.

通讯控制 I-变量

I1: 串口模式

= 0, 2 :Host CS hand shake used

= 1, 3 :Host CS hand shake disabled

= 2, 3: Multiple card daisy chain enabled

I3: 握手字符控制

= 1: <LF> sent to acknowledge valid command

= 2, 3: <ACK> sent to acknowledge valid command

= 1, 3: <LF> precedes data response line

I4: 通讯完整性模式

= 0, 2 :Checksum disabled

= 1, 3: Checksum enabled

= 0,1 : Serial errors reported immediately

= 2, 3: Serial errors reported at end of line

I6: 错误报告模式

= 0, 2: Returns <BELL> character

= 1, 3: Reports error number

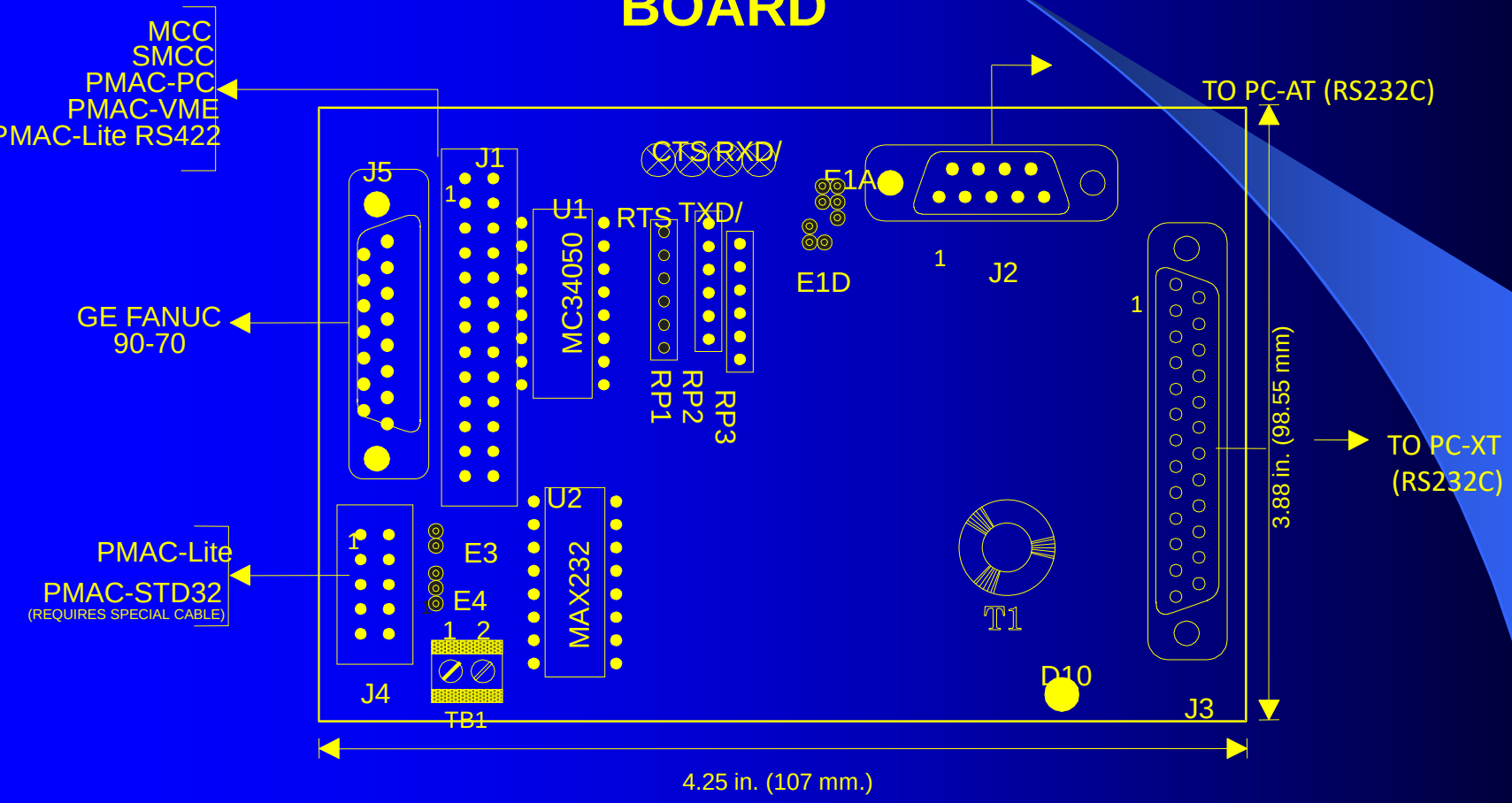
Communications Control I-Variables

I9: **Data Reporting Form**
= 1, 3: Long form data reporting
= 2, 3: Hexadecimal address I-variable reporting

I58: **DPRAM ASCII Communications**
= 0: Disables DPRAM communication
= 1: Enables DPRAM communication

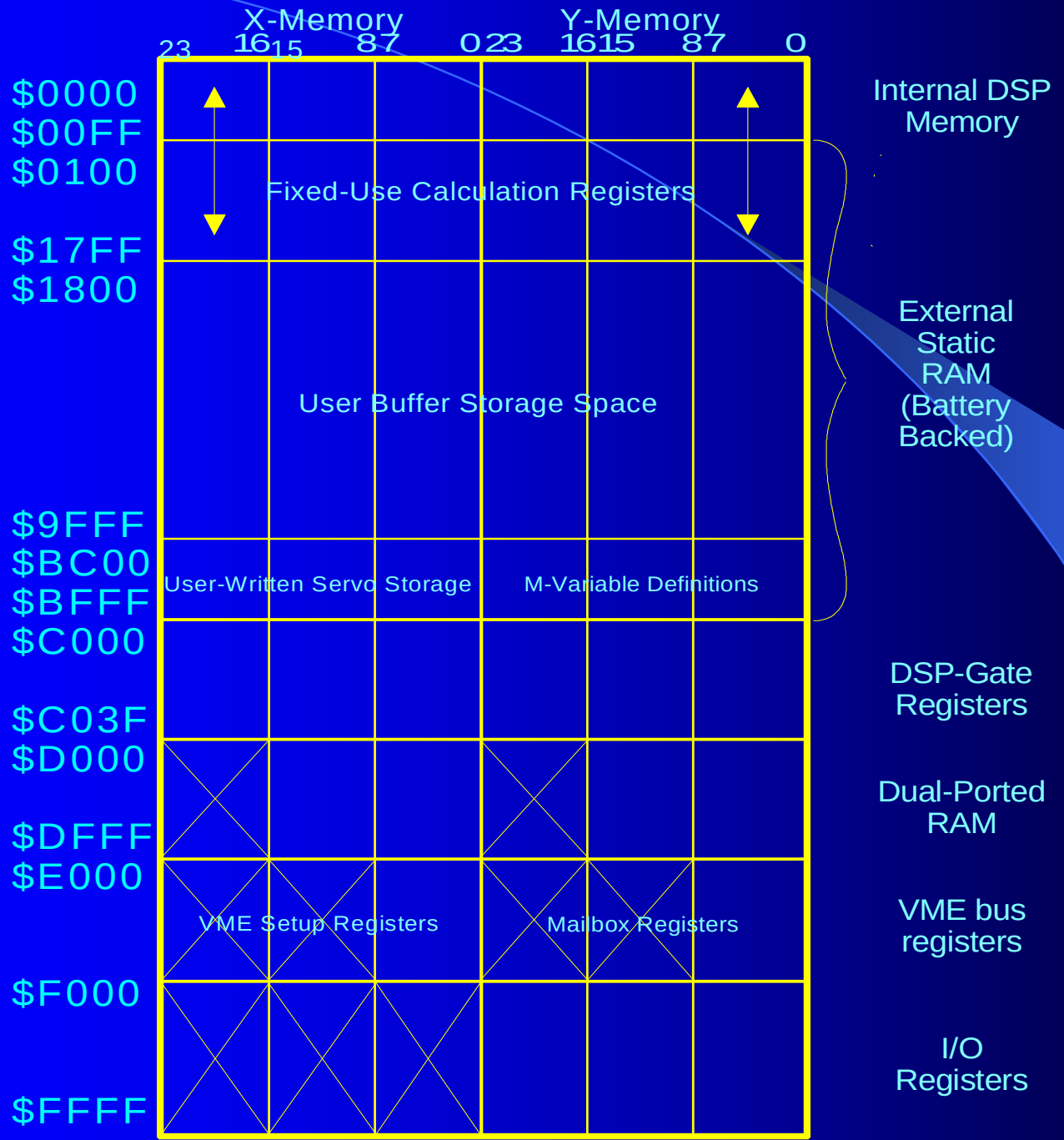
I56: **DPRAM ASCII Communications Interrupt Enable**
= 0: Disables communications interrupt
= 1: Enables communications interrupt

ACC-26A
RS422/232 TO RS232
OPTICALLY ISOLATED CONVERTER BOARD



FOR RS-422 INSTALL U1, RP1, RP2, RP3, REMOVE U2
FOR RS-232 INSTALL U2, REMOVE U1, RP1, RP2, & RP3

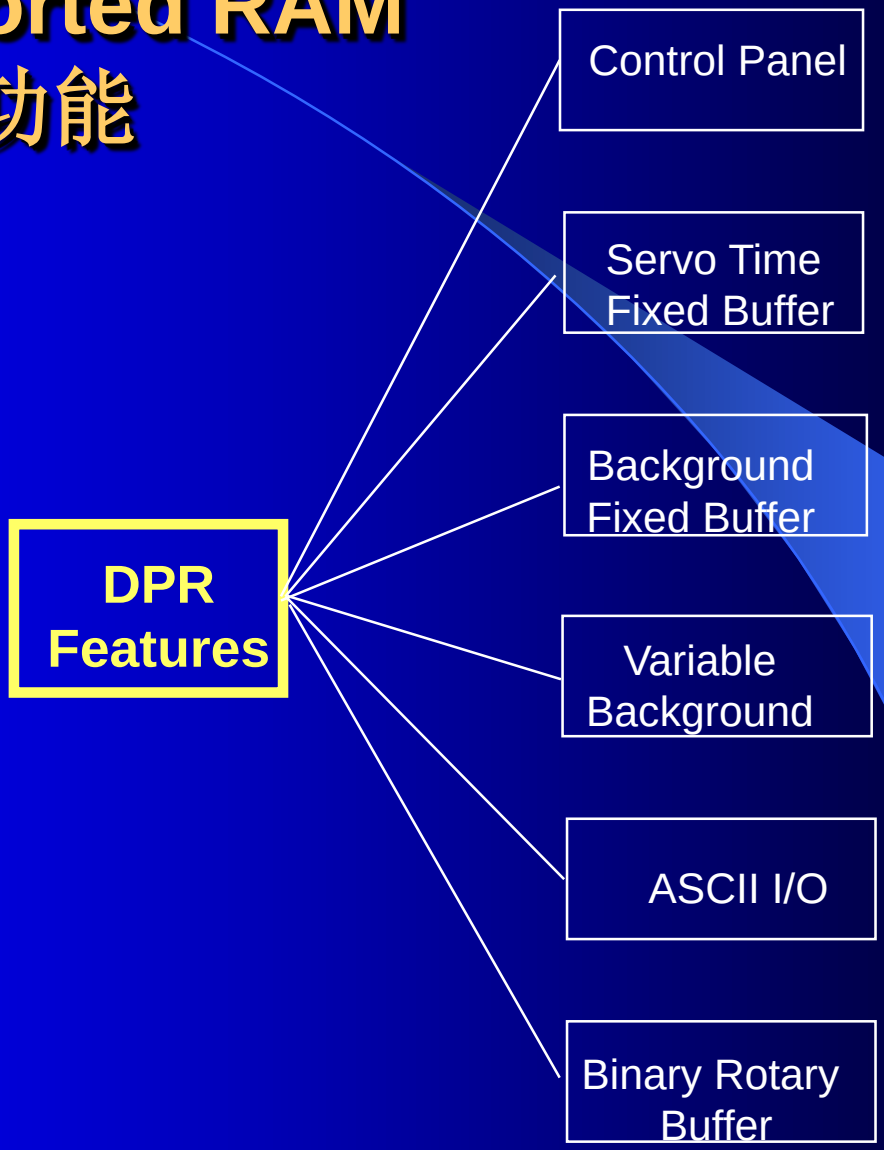
PMAC Memory Mapping



PMAC Dual-Ported RAM 内存图

Host Address Offset		PMAC Address
0x000	Control Panel Functions	$D000
0x0024	Servo Data Reporting Buffer	$D009
0x0228	Background Data Reporting Buffer	$D08A
0x062C	ASCII Command Buffer (HOST to PMAC)	$D18B
0x06D0	ASCII Response Buffer (PMAC to Host)	$D1B4
0x7E8	Pointers to Variable Size Buffers	$D1FA
0x0800	Room for Variable-Size Buffers (1) Data Gathering (2) Background Variable Data (3) Binary Rotary Program	$D200
0x3FFC	Open Use Space	$DFFF

PCOMM Dual Ported RAM
特性支持功能



PMAC 通讯软件

Name	Description	16-bit	32-bit
ACC-9P	PCOMM	Yes	No
ACC-9PN	PCOMM-32	No	Yes
ACC-9PT	PTALK OCX	No	Yes
ACC-9W	PMAC Executive for Windows	Yes	No
ACC-9WN	PMAC Executive for Windows 95,98, or NT	No	Yes
ACC-9DA	PMAC Executive for DOS	Yes	No
ACC-33W	PMAC NC for Windows	Yes	No
ACC-33WN	PMAC NC for Windows 95,98, or NT	No	Yes

新的PMAC 软件

PMAC Panel - VI’s for LABView

PMAC 连通性 - OPC Server for different field busse ControlNet, DH+, DeviceNet, Ethernet